Carbonatite-hosted REE prospectivity in Canada from national to provincial scale: leakage-free validation, transferability and a semi-supervised deviation network under extreme label scarcity

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Abstract

Carbonatite-hosted rare earth element (REE) prospectivity models must learn from a few tens of occurrences and no confirmed barren ground. We evaluate them at two scales within one framework: the same occurrence table, the same input rule and the same validation for Canada as a whole and for the Superior Province (15 occurrences), with the Cordilleran Orogen and Grenville Province as contrasts. Input values that depend on the extent of the study area (raster clips, texture grey levels, generalised geology) are computed per geological province, so a point has the same value at both scales. Rebuilding the national S2-grid design of an earlier study reproduces its test AUC of 1.00; child-cell leakage and negatives drawn from the least similar ground each suffice for it, and with both removed the same random forest reaches an average precision (AP) of 0.18 at realistic prevalence and 0.03 under 100 km spatial blocks. Nationally, the best AP under 100 km blocks is 0.06 (random ranking: 0.003); models transfer between provinces mostly at chance level, and pooling the country into a province's training data lowers that province's blocked AP in most cases. Within Superior, a semi-supervised Deviation Network (DevNet) has an AP of 0.45 under random cross-validation and 0.42–0.25 under 50–200 km blocks, ahead of random forest, XGBoost, bagging-PU and nnPU (random forest and XGBoost: 0.27–0.28 and 0.12–0.20), although with 15 occurrences its margin is not statistically conclusive. Nationally DevNet is weak: giving each province its own background reference doubles its AP, which locates the deviation loss's assumption of one normal background at the scale of a province. An ablation that labels the background barren

matches DevNet within Superior; treating the background as unlabelled pays off when point-level map classes are withheld.

Keywords: Mineral prospectivity mapping, Rare earth elements, Carbonatite, Spatial scale, Transferability, Deviation network, Spatial cross-validation, Data leakage

1 Introduction

Rare earth elements (REEs) are critical to low-carbon energy, electronics and defence technologies, yet economic concentrations are geologically uncommon (Wang, Fan, Zhou, Yang, & She, 2020). Most REE resources are hosted by carbonatites and associated alkaline intrusions, whose distinctive but partly overlapping geophysical and geochemical signatures make them natural targets for data-driven mineral prospectivity mapping (MPM; Simandl & Paradis, 2018; Verplanck, Van Gosen, Seal II, & McCafferty, 2014). In Canada, however, confirmed carbonatite-hosted REE occurrences are few and spatially clustered (Fig. 2): after near-duplicates are merged, a single geological province holds between about ten and thirty of them.

Two properties of this problem undermine standard supervised classification. First, positives are extremely scarce: mineralisation is a rare event (Xiong & Zuo, 2018), known occurrences are only a partial and biased sample of the mineral system (Coolbaugh, Raines, & Zehner, 2007; Parsa & Carranza, 2021; Zuo et al., 2021), and under k-fold cross-validation each fold holds only a handful of them, so any estimate is dominated by which occurrences happen to be held out. Second, there are no true negatives. Ground without a recorded occurrence is unexplored rather than barren, and designating it negative is an arbitrary modelling choice whose outcome depends on which background points are drawn (Zuo & Wang, 2020). Together these properties motivate positive–unlabelled (PU) and semi-supervised anomaly-detection formulations, in which mineralised sites are treated as rare anomalies relative to a large, mostly barren but unlabelled population (Bekker & Davis, 2020; Y. Chen & Wu, 2019; Luo, Farahbakhsh, Hore, & Müller, 2026; Xiong & Zuo, 2016, 2021).

The Deviation Network (DevNet; Pang, Shen, & van den Hengel, 2019) is a semi-supervised deep anomaly detector that learns an anomaly score from a few labelled anomalies and a large unlabelled pool, pulling the scores of unlabelled data towards a Gaussian reference while pushing labelled anomalies a fixed margin above it. Luo et al. (2026) adapted it to MPM (DEEP-SEAM), encoding geology as distances to mapped units and adding neighbourhood statistics of the evidence layers. Whether such a model outperforms well-tuned discriminative baselines when positives number in the low tens has not been tested under evaluation designs that control for the specific ways in which small, clustered label sets inflate scores, nor has it been separated from the simpler explanation that any smooth neural scorer would do as well. The closest prior study, a national-scale model of Canadian carbonatite-hosted REE ± Nb deposits (Parsa et al., 2024), reported test AUCs of 0.91–0.96 with random forests and one-dimensional convolutional networks. Its labels were augmented by replacing each occupied S2 grid cell with its four child cells, its negatives were drawn from the ground

least similar to the positives, and it was evaluated by stratified random 70/30 splits of the augmented labels. A follow-up study recommended synthetic labels generated by generative adversarial networks (GANs) for the same commodity (Parsa et al., 2025). Each of these design choices can make the held-out test easier than prospecting new ground. A national model also answers a different question from a provincial one. Pooling every occurrence of a country raises the label count, but it mixes occurrences from geological provinces whose geophysical settings, and whose barren ground, differ completely; whether a model trained on such a pool predicts within any one province, and whether one province's occurrences say anything about another's, is rarely tested (Meyer & Pebesma, 2021; Ploton et al., 2020).

This study evaluates carbonatite-hosted REE prospectivity models at two scales: Canada as a whole and the Superior Province of the Canadian Shield, the province in which labels are scarcest relative to area, with the Cordilleran Orogen and Grenville Province modelled with the identical pipeline as contrasting cases. For differences between the scales to be attributable to scale, both use one occurrence table, one input rule and one validation design; in particular, every input whose value depends on the extent of the study area is computed per geological province, so that a point has the same value in the national and the provincial analysis. We ask: (i) how well can occurrences be predicted nationally once leakage is removed from validation, and how much of the high AUC of national S2-grid designs does the design itself produce? (ii) do models transfer between provinces, and does pooling the country help a province? (iii) within Superior, does DevNet outperform tree-based learners, including tuned ensembles and XGBoost, and PU learners once every model sees the same data and folds, and does any advantage survive spatially blocked validation? (iv) which of DevNet's components produce any advantage, when does treating unexplored ground as unlabelled rather than barren pay off, and at what scale does its premise of one normal background hold? and (v) how much of the skill comes from geological map classes that mark known complexes, and can synthetic positives compensate for label scarcity? We close by interpreting the provincial models with SHapley Additive exPlanations (SHAP), using a rank-based rule for the direction of each feature's effect, and by checking the Superior maps against an untrained REE resource and the mapped Kapuskasing Structural Zone.

2 Geological setting and data

2.1 Carbonatite-hosted REE systems and their geophysical expression

Carbonatites occur predominantly in Precambrian cratonic areas, and individual provinces record repeated emplacement over hundreds of millions of years, implying long-lived, episodically reactivated trans-lithospheric pathways (Woolley & Bailey, 2012). A component of their REE endowment is attributed to subcontinental lithospheric mantle refertilised by REE- and CO_2-rich metasomatic fluids (Hou et al., 2023; Yaxley, Kjarsgaard, & Jaques, 2021). Each element of this source–transport–trap system has a mappable geophysical proxy (Fig. 1). Depth to the lithosphere–asthenosphere boundary (LAB) indexes the thickness and edge geometry of the

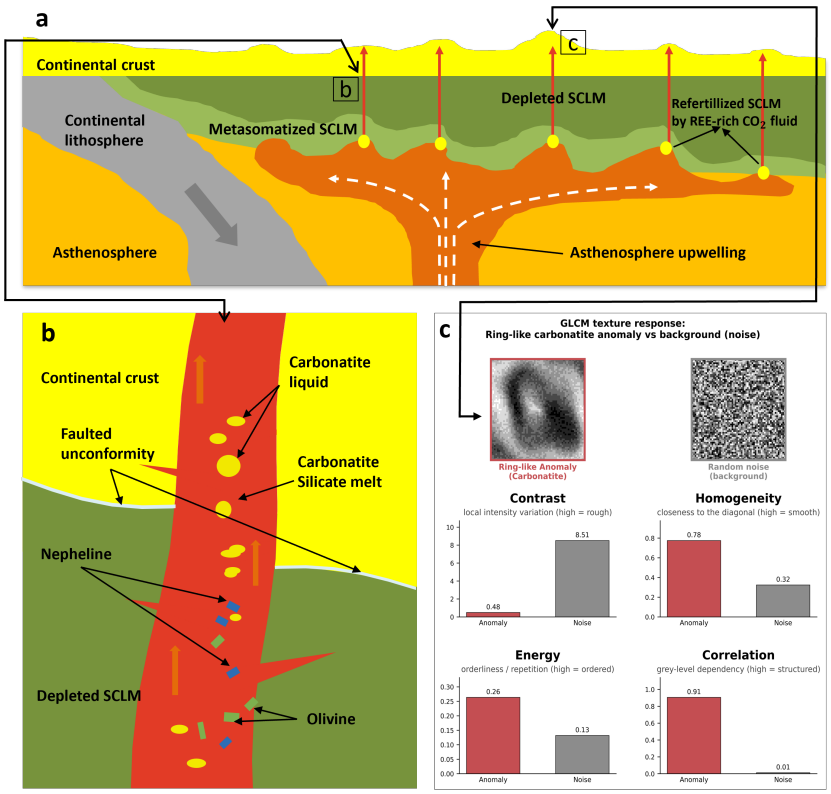


Fig. 1 Source–transport–trap model for carbonatite-hosted REE systems and its geophysical proxies. (a) Source: refertilised subcontinental lithospheric mantle near the LAB at the craton margin, indexed by LAB depth and margin proximity; panel modified after Hou et al. (2023). (b) Transport along trans-lithospheric faults, indexed by fault and worm proximity and Moho depth; modified after Yaxley et al. (2021). (c) Trap: GLCM texture response of a ring-like anomaly and of random noise.

cratonic root that hosts the fertile mantle source, which exerts first-order control on magmatic ore systems (Griffin, Begg, & O'Reilly, 2013; Groves & Santosh, 2021; Hoggard et al., 2020); Moho depth indexes crustal thickness; and proximity to faults and to the loci of maximum horizontal gravity and magnetic gradient ("worms"; Hornby, Boschetti, & Horowitz, 1999) proxies the structures that channel ascending melts. At the emplacement site, dense and magnetic alkaline–carbonatite complexes produce positive gravity anomalies and intense circular to annular magnetic anomalies (Drenth, 2014; Thomas, Ford, & Keating, 2016). Grey-level co-occurrence matrix (GLCM) textures summarise the local two-dimensional heterogeneity produced by such concentric patterns without assuming a target diameter (Cooper & Cowan, 2005; Hall-Beyer, 2017), unlike explicit radial-symmetry detectors tuned to ring-complex size (Aranha, Porwal, Sundaralingam, & González-Álvarez, 2021).

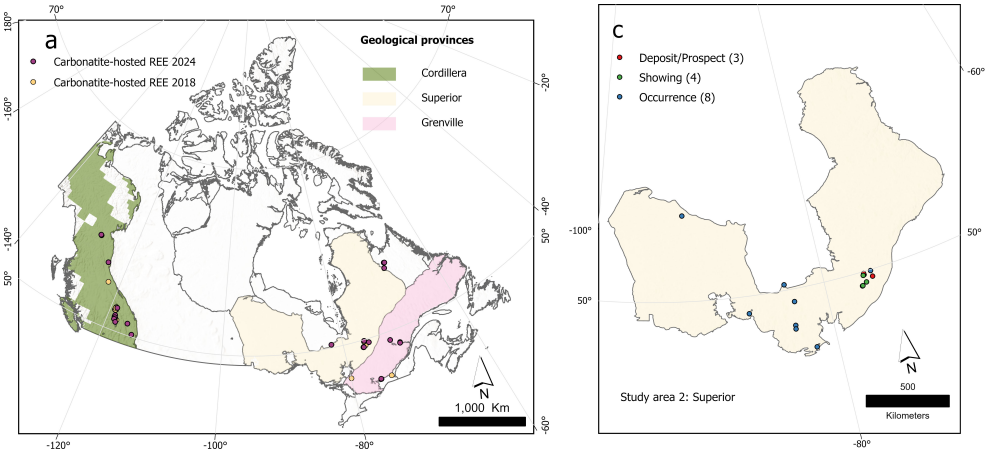


Fig. 2 (a) Carbonatite-hosted REE occurrences in Canada and the three modelled provinces; coloured areas are the sampleable domains. (b) The 15 deduplicated Superior occurrences used as positives, by development class.

2.2 Study areas

The national study area is the union of the onshore geological provinces, orogens and platforms of the Geological Map of Canada (Map 1860A; Wheeler et al., 1996), without the continental shelves and oceanic crust (12.3 million km^2). The three provincial study areas are the Superior Province, the Grenville Province and the Cordilleran Orogen polygons of the same map. Carbonatite-hosted occurrences of the Phanerozoic, accreted Cordillera are geophysically distinct from those of the Canadian Shield (Parsa et al., 2024), which suggests that a single cross-terrane model would not transfer; Section 4.1 tests this directly. Windows of one province enclosed by another and smaller than 2,000 km^2, such as the Churchill-basement windows of the Monashee and Malton complexes in the Cordillera, were assigned to the surrounding province.

The Superior Province is the world's largest Archean craton. Carbonatite–alkaline magmatism in the Canadian Shield clusters into episodes at about 2.68 Ga, 1.90–1.88 Ga and 1.16–1.08 Ga (Rukhlov & Bell, 2010), and in the Superior Province many of the complexes are aligned along crustal-scale structures, most prominently the Kapuskasing Structural Zone (KSZ; Gittins, Macintyre, & York, 1967; Percival & West, 1994). The 15 Superior occurrences used here (Fig. 2b) comprise a cluster around the Montviel and Lac Shortt carbonatites in the eastern Abitibi region of Québec (Deposit/Prospect and Showing class) and occurrences dispersed across the southern and western Superior Province, including Nemegosenda Lake, Lackner Lake, Kapuskasing, Martison Lake, Prairie Lake and Carb Lake. The Superior Province thus combines the fewest occurrences per unit area of the three provinces with a spatial structure, one tight cluster plus scattered singletons, in which a model can either learn the cluster or learn what the scattered occurrences have in common.

2.3 Occurrence labels

Positive labels were compiled from the carbonatite-hosted records of the USGS global REE occurrence database (Orris, Seo, Briggs, Dunlap, & Cocker, 2018) and the occurrence table of Parsa et al. (2024). Alkaline-hosted occurrences were excluded after a preliminary leave-block-out classification showed them to be geophysically separable from carbonatite-hosted ones. Records within 2 km of each other were merged by single linkage, so that drill holes or zones of one deposit, listed once or in both sources, count once; without this step near-identical points from one deposit fall into both training and validation folds. Merging and the data-coverage filter (Section 2.6) reduced 21, 13 and 51 pooled records to 15 (Superior), 12 (Grenville) and 27 (Cordillera) positives. Each positive keeps the development class of Parsa et al. (2024): Deposit/Prospect, Showing or Occurrence (Superior: 3, 4 and 8). The national label table takes the three provincial tables as they are and adds the 13 occurrences elsewhere in Canada, merged by the same rule (five in the Churchill Province, three in the St. Lawrence Platform, two in the Hudson Bay Lowlands, two in parts of the Cordilleran Orogen outside the study polygon and one in the Slave Province); two records outside a study polygon but within 2 km of one of its occurrences were merged into that occurrence. After the data-coverage filter the national table holds 66 occurrences (Fig. 3).

2.4 Geophysical and structural evidence layers

Table 1 lists the evidence layers. Seventeen national rasters cover the trap and the deep-architecture components: Bouguer gravity and its derivatives and filtered versions, total magnetic intensity and its differential reduction to the pole with derivatives (Jobin, Véronneau, & Miles, 2017; Miles & Oneschuk, 2016), together with long-wavelength gravity and magnetics, LAB and Moho depth from the Tri-National Critical Minerals Mapping Initiative release (McCafferty et al., 2023). Eleven of the gravity and magnetic grids are processed products supplied by Getech Group plc; four of these are gridded versions of the public Natural Resources Canada compilations and seven are standard derivatives of them (Table 1). Two line layers (ancient passive margins, D.C. Bradley, 2008; faults) and four point layers (shallow and deep gravity and magnetic worms; McCafferty et al., 2023) enter as normalised proximities. GLCM contrast, homogeneity, energy and correlation were computed in 9×9-pixel windows on the Bouguer anomaly, its 40 km high-pass, the reduced-to-pole magnetics and the total magnetic intensity. Following Luo et al. (2026), the mean and standard deviation of the same four grids within 10 km of each point were added as neighbourhood statistics. Radiometric grids (K, eU, eTh) were considered but not used, because their coverage of the study areas is incomplete and uneven; whole-rock geochemistry is too sparse to give a value at every map cell. The province polygons of Wheeler et al. (1996) define the study areas only; no distance-to-province-boundary feature was used.

2.5 Geology

Geology enters from a hexagonal H3 grid (resolution 7, cells of about 5 km^2) attributed from the Geological Map of Canada (Wheeler et al., 1996). Each cell carries a major

Table 1 Evidence layers, grouped by mineral-system component. Native resolution in brackets. All rasters were sampled at points in their native projection and resolution.

Component	Layer(s)	Source
Source / lithosphere	Depth to LAB (0.1°); Moho depth (0.25°)	McCafferty et al. (2023)
Deep architecture	Bouguer gravity upward-continued 30 km and its horizontal-gradient magnitude (HGM) (0.031°); long-wavelength (deep-source) RTP magnetics and its HGM (0.02°)	McCafferty et al. (2023)
Transport	Proximity to faults; to ancient passive margins; to shallow and deep gravity and magnetic worms	McCafferty et al. (2023); D.C. Bradley (2008)
Trap: gravity	Bouguer anomaly (BA), BA first vertical derivative, BA total horizontal derivative (THD) (2 km)	Jobin et al. (2017); gridded by Getech
	BA THD tilt, BA 40 km and 100 km high-pass (4th-order Butterworth) (2 km)	Getech derivatives
Trap: magnetics	Total magnetic intensity (TMI) (0.01°)	Miles and Oneschuk (2016); gridded by Getech
	Differential reduction to pole (DRTP), DRTP 1VD, DRTP THD, DRTP THD tilt (0.01°)	Getech derivatives
Anomaly form	GLCM contrast, homogeneity, energy, correlation (9×9 window) on BA, BA 40 km high-pass, DRTP, TMI; mean and standard deviation within 10 km of the same grids	this study
Geology	H3 cell (resolution 7): major lithology class, era, lithological- and age-contact flags; distances to contacts and units of a generalised map (Section 2.5)	Wheeler et al. (1996)

lithology class (for example Igneous Intrusive Alkalic or Metamorphic Gneiss Paragneiss), its dominant era, and flags for cells crossed by a lithological or an age (period) contact; each point takes the attributes of the cell that contains it. The lithology class enters as one-hot columns (a class gets its own column if at least 10 of the region's 5,000 training-background points fall in it; rarer classes share one), together with the era one-hot and the two contact flags. Following Luo et al. (2026), geology also enters as distances (km) to six elements of a generalised map, from which every lithological unit smaller than 20 connected cells (about 100 km^2) and the alkalic-intrusion class were removed: lithological contacts, age contacts, greenstone (extrusive) units, mafic–ultramafic intrusions, high-grade metamorphic units and sedimentary units. After pruning (Section 3.2) the inputs number 48 in Superior (24 geophysical and structural layers, 3 neighbourhood statistics, 2 regional distances, 15 lithology columns, 2 era columns and 2 contact flags), 55 in Cordillera and 58 in Grenville.

At the scale of a carbonatite complex a bedrock map can mark the target itself: in Superior, 6 of the 15 occurrences fall on cells mapped as alkalic intrusive rock, against 16 of the 3,000 evaluation-background points (0.5%). A mapped alkaline complex is legitimate evidence for an explorer, but a model can score well simply by looking it up. We therefore also report the models without the point-level map classes (the lithology, era and contact columns), keeping the geophysical and structural layers, neighbourhood statistics and regional distances (29, 31 and 33 inputs after pruning). This measures what the models achieve on ground where the map does not show a complex.

2.6 Sampleable domain and background

Rasters were clipped to each study-area outline buffered outward by one GLCM window radius (8 km), so that windows near the boundary read real data, and sampling and prediction were restricted to the domain where every layer holds data and every texture window is computable. This domain covers 98.6% of the Superior, 97.2% of the Grenville and 73.5% of the Cordilleran outline and 8.09 million km^2 (65.6%) of Canada; one Cordilleran occurrence (Lempriere) and one St. Lawrence Platform occurrence (Francon Quarry) fall outside it and were dropped. Background candidates are the nodes of a 0.1° grid inside the domain and more than 10 km from any known occurrence (19,371 in Superior). From these, two disjoint pools were drawn once per region at random: an evaluation background of 3,000 points and a training background of 5,000 points. Every model in this study is trained on the positives of its training fold plus the same 5,000 training-background points and scored on the held-out positives plus the same evaluation background. The prevalence in the scored pool is therefore 15/3,015 $\approx$ 0.5% in Superior, which is the AP of a random ranking.

The national pools follow the same rule at Superior's density, so that a national point inside a study province is always also a provincial one. The provincial pools hold 41% of Superior's background candidates; inside Superior the national pools are the provincial ones, inside Cordillera and Grenville they are random subsets of the provincial pools of 41% of each province's candidates, and elsewhere new pools of 41% of the candidates were drawn, with evaluation and training background split 3:5 as in each province. National points must also lie on the H3 geology map. The national table holds 66 occurrences, 20,855 evaluation-background and 34,761 training-background points (prevalence 0.32%; Fig. 3).

3 Methods

Figure 4 summarises the workflow. Code for every analysis, including the exact numbers in every table, is available in the repository listed under Code availability.

3.1 Two scales, one framework

Three inputs depend on the extent of the area over which they are computed: raster values near the edge of a clip (a coarse pixel whose centre falls outside the buffered outline is nodata), the GLCM grey levels, which are quantised between the 2nd and 98th percentiles of the clipped raster, and the generalised geology map, which is built from the H3 cells in the outline's bounding box plus a margin (2° of latitude and 3° of longitude). Computed over the whole country, the same point takes a different texture value than in its province (for the Bouguer GLCM correlation in Superior the rank correlation between the two is -0.002). All three are therefore computed per geological province, with each province's own buffered clip and bounding box, and the study provinces use exactly the provincial rasters and boxes; inside them every national value equals the provincial one, which we verified column by column. Appendix B, Table B4, shows what computing them over the country instead would have cost. Feature pruning (Section 3.2) and the H3 class list are applied to the table

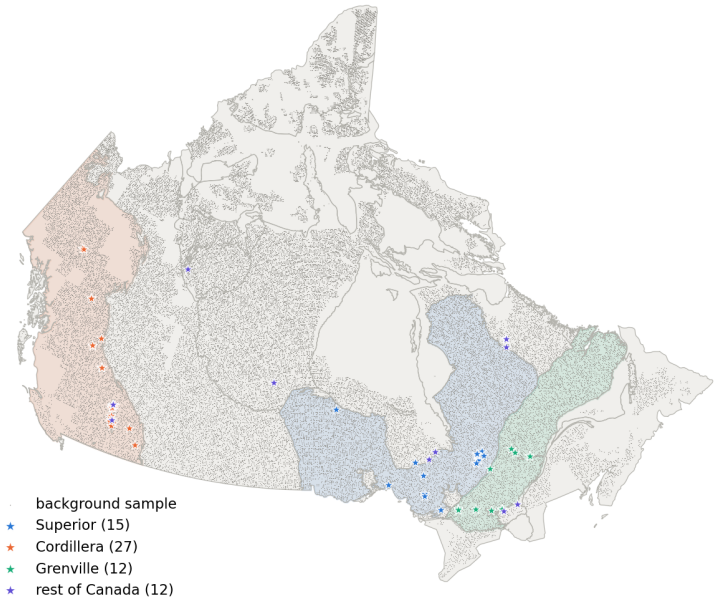


Fig. 3 The national table: 66 carbonatite-hosted REE occurrences by province group and the background sample (grey dots; 55,616 points) on the onshore geological provinces of Map 1860A (Wheeler et al., 1996); the three study provinces are shaded. Inside the study provinces the points are those of the provincial analyses. The background is denser at high latitudes because the 0.1° lattice converges northwards.

on which a model is trained: national models use the 65 inputs pruned on the national table (all 26 lithology classes have at least 10 national training-background points), and models of one province use that province's inputs and network capacity, also when they are trained on, or applied to, other provinces' points.

3.2 Feature pruning

The candidate features include many collinear geophysical derivatives. Within each region, features were clustered hierarchically on $1-|\rho_S|$ (Spearman correlation, average linkage) and cut at $|\rho_S| = 0.7$, keeping one representative per cluster chosen by lowest missing rate and then highest variance (Dormann et al., 2013). The rule never sees labels and therefore cannot leak information across folds.

3.3 Models

All models see the same rows and differ only in how they treat the training background or in their loss.

Random Forest (RF; Breiman, 2001) is the discriminative baseline: positives versus training background labelled 0, 300 trees, $\sqrt{p}$ features per split, balanced class weights, inside a pipeline that fits median imputation and standardisation on the

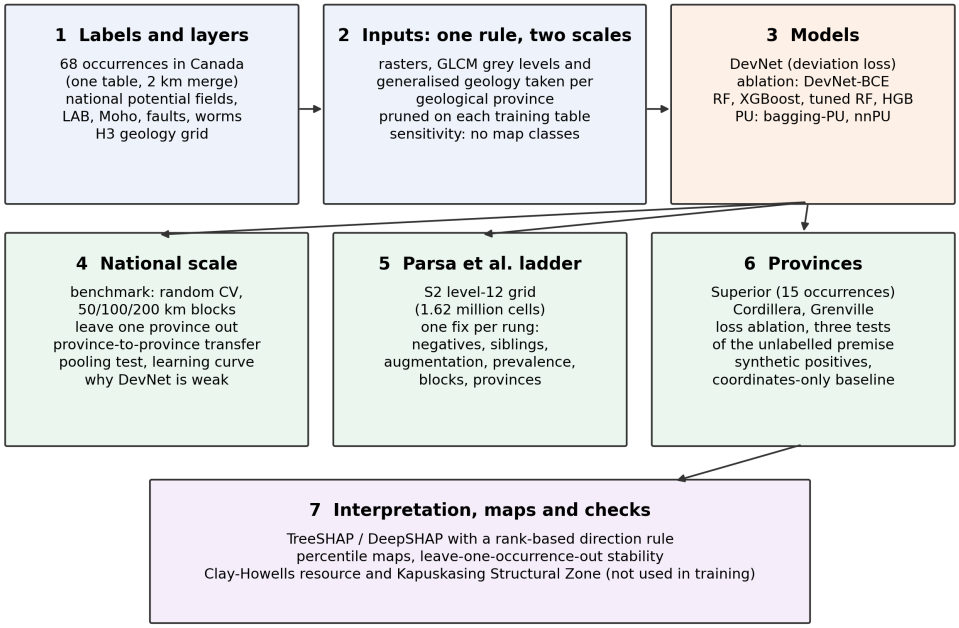


Fig. 4 Workflow. One occurrence table, one input rule and one validation design serve both scales. Within a comparison, all models are trained on the same positives and background and scored on the same folds; only the treatment of the background, or the loss, differs.

training fold only. With roughly 12 training positives against 5,000 background points, most held-out points receive a predicted probability of exactly zero, so the pooled AUC depends on which trees happen to vote for a given held-out occurrence. We therefore vary the RF seed across folds so that this variance enters the reported intervals.

XGBoost (T. Chen & Guestrin, 2016) adds a gradient-boosted tree ensemble with parameters fixed in advance from a parallel national study of the same commodity rather than tuned here: 200 trees of depth 2, learning rate 0.1, row and column subsampling of 0.8, minimum child weight 3 and L2 and L1 penalties of 2 and 0.5. It is trained like RF, on median-imputed inputs with the positives up-weighted to balance the two classes.

DevNet (Pang et al., 2019) maps each input vector $\mathbf{x}$ to a scalar score $\phi(\mathbf{x})$ through one hidden ReLU layer with dropout. The deviation $d(\mathbf{x}) = (\phi(\mathbf{x}) - \mu_R)/\sigma_R$ is measured against the mean and standard deviation of a fixed reference sample of 50,000 standard-normal draws, and the loss

$$L = (1 - y)\,|d(\mathbf{x})| + y\,\max\{0,\, a - d(\mathbf{x})\} \tag{1}$$

pulls unlabelled background ($y = 0$) towards the reference and pushes positives ($y = 1$) beyond a margin $a = 5$, the value of Pang et al. (2019) (Appendix F, Table F12). Network capacity was set a priori by positive count (64 hidden units, dropout 0.3,

weight decay 10^{-4} in Cordillera; 32, 0.4 and 10^{-3} in Superior and Grenville) rather than tuned, because a search over a few tens of positives fits noise. Training uses Adam (Kingma & Ba, 2014) with weight decay, cosine learning-rate annealing over 300 steps and gradient clipping, and each fold score is the rank average of five restarts with different seeds.

Two design choices adapt DevNet to tabular evidence layers with very few positives. First, inputs are mapped to standard-normal marginals by a quantile transformation fitted on each training fold. Many evidence layers are strongly skewed (inverted distances concentrated near 0 or 1, long-tailed textures, 0/1 dummies), and unlike tree ensembles, which are invariant to monotone transformations, multilayer perceptrons are sensitive to such marginal distributions (Gorishniy, Rubachev, Khrulkov, & Babenko, 2021; Grinsztajn, Oyallon, & Varoquaux, 2022). Second, each gradient step uses 16 positives (resampled with replacement) and 240 background rows. The deviation loss is meant to learn what the background looks like; with label-balanced batches of 16 + 16 rows, 300 steps would show each background row about once and each positive about 400 times, whereas background-heavy batches show each background row about 14 times while positive exposure is unchanged. Both choices were settled with diagnostic runs on these provinces using an earlier, coarser encoding of the geology; Appendix F repeats the ablation with the inputs used here.

Bagging-PU (Mordelet & Vert, 2014) trains 200 fully grown decision trees, each on all training positives and an equal-sized bootstrap sample of the unlabelled background, and averages their outputs. It is the standard tree-based PU learner and the closest PU analogue of RF.

nnPU (Kiryo, Niu, du Plessis, & Sugiyama, 2017) minimises the non-negative PU risk with the sigmoid loss, using exactly DevNet's network, quantile-normal inputs, optimiser, number of steps and five-restart ensemble, so that DevNet versus nnPU isolates the PU risk from the deviation loss. nnPU requires the class prior $\pi = P(y = 1)$ in the unlabelled population; we use $\pi = 0.01$ (sensitivity in Appendix A, Table A3). The calibration approach of Elkan and Noto (2008) is not included because under its assumptions the calibrated output is a monotone transform of the positive-versus-background classifier, so its ranking, and every rank metric reported here, equals RF's.

RF-tuned and HGB-tuned give the tree-based baselines a design search of their own. Inside every outer training fold, an inner spatially blocked cross-validation (100 km blocks, three folds, training background purged from inner validation blocks) chooses the RF's minimum leaf size (1 or 10) and number of features per split ($\sqrt{p}$ or $p/3$), or the number of leaves (8 or 31) and minimum leaf size (5 or 40) of a histogram gradient-boosting classifier (Pedregosa et al., 2011) with balanced class weights, 200 iterations, learning rate 0.05 and an L2 penalty of 1. The outer test fold is never used for the choice.

3.4 Evaluation protocol

With three or four positives per validation fold, per-fold AUC is unstable and can be undefined, so out-of-fold scores are pooled (A.P. Bradley, 1997): each held-out point is scored once per repeat, scores are converted to within-fold percentile ranks

so that folds are comparable, and a single AUC and average precision (AP) are computed over all folds. Pooling carries a small known negative bias in AUC (Airola et al., 2019; Parker, Günter, & Bedo, 2007). We emphasise AP because, at 0.5% prevalence, it reflects the practical task of concentrating occurrences into a small search area, whereas AUC is dominated by the bulk of easily rejected background (Saito & Rehmsmeier, 2015). Exploration efficiency is also shown as capture-rate curves: the share of held-out occurrences ranked within a given top fraction of the scored pool (Agterberg & Bonham-Carter, 2005; Chung & Fabbri, 2003). Because the evaluation background is a random sample of the domain, this fraction approximates search area. Because unexplored background may host undiscovered occurrences, all metrics are conservative.

Random cross-validation. Repeated stratified 5-fold cross-validation with six repeats gives six pooled AUC and AP values per model, reported as the median and the 2.5–97.5 percentile interval over repeats (labelled 95% interval). All models use identical splits, so they can be compared pairwise.

Spatially blocked cross-validation. Random folds can hold out an occurrence a few kilometres from a training occurrence. Following Roberts et al. (2017) and Valavi, Elith, Lahoz-Monfort, and Guillera-Arroita (2019), labelled points were assigned to square blocks of 50, 100 or 200 km side in Canada Atlas Lambert (EPSG:3978), whole blocks were held out together using stratified group 5-fold splitting (fewer folds if fewer blocks contain positives), and every training-background point falling inside a held-out block was removed from training for that fold. Each of six repeats uses a random offset of the block grid.

Paired comparisons. For each pair of models we average each point's out-of-fold rank over the six repeats and draw 5,000 stratified bootstrap resamples of positives and background (Efron & Tibshirani, 1993), reporting the difference in AP and AUC with its 95% percentile interval and the number of the six repeats in which the first model has the higher AP. With 12–27 positives these comparisons have little power, so we treat a difference as established only when it is consistent across repeats, block sizes and provinces.

Geography-only baseline. To separate geological signal from spatial memorisation, an RF was trained on the projected coordinates of each point only, with no evidence layer, under the same folds and both protocols. Its score measures how much of a protocol's reported skill can be obtained simply by remembering where the training occurrences lie. An L2-regularised logistic regression with balanced class weights serves as a smooth, essentially linear discriminative reference.

3.5 National experiments

Benchmark. Every model was evaluated on the national table with the protocols of Section 3.4: random 5-fold cross-validation with six repeats and 50, 100 and 200 km blocks. Out-of-fold scores were also evaluated on the points of each study province alone.

Transfer between provinces. In leave-one-province-out validation a model trained on the national table without one province group (Superior, Cordillera,

Grenville or the rest of Canada) scores that group's occurrences and evaluation background. In province-to-province transfer, the model of province A (its inputs, capacity, occurrences and training background) scores the occurrences and evaluation background of province B; the diagonal, random cross-validation within A, is the provincial result. Each fit was repeated with six seeds.

Pooling. For each study province the same held-out occurrences, evaluation background and folds (random cross-validation and 100 km blocks) were scored by four versions of the province's model, trained on the province's own data, on these plus the national training background outside the province, plus the national occurrences outside it, or plus both, the last being a national model in the province's specification. Under blocks, added points that fall in a held-out block were removed as the province's own background is.

Province differences and sample size. A random forest (300 trees, 5-fold cross-validation) tried to tell the training background of one province group from another's, and the occurrences of one province from another's; an AUC of 0.5 means indistinguishable and 1.0 fully separable distributions. For each input, the AUC of the input alone for a province's occurrences against its evaluation background gives the direction in which it marks occurrences there. A learning curve kept $k = 3, 5, 10, 20$ or 40 training occurrences per fold (or all) under national random cross-validation.

Why DevNet is weak nationally. Before running them we wrote down one hypothesis and three predictions, each judged against RF as a control. The hypothesis: DevNet scores a point by its deviation from the training background, its reference of "normal", and the national background mixes provinces whose distributions are separable, so the reference is not concentrated. P1 (offsets): ranking DevNet's national out-of-fold scores within each province group raises its AP clearly more than RF's. P2 (local reference): one model per province group, trained on all training occurrences against that group's background only and scoring that group's points, raises DevNet's AP more than RF's. P3 (indicator): adding the province group as one-hot inputs helps DevNet more than RF. Under-training was checked separately by scaling DevNet's 300 steps with the size of the background.

3.6 The national S2-grid design of Parsa et al. (2024), corrected rung by rung

We rebuilt the national S2-grid design with the same geometry library on every level-12 S2 cell (about 4.5 km^2) whose centre lies in the national domain (1,616,693 cells), with the national inputs extracted at the cell centres. The 66 occurrences fall in 66 cells. Each rung changes one element of the design relative to the one before (Table 2). R0 follows Parsa et al. (2024): each occurrence cell is replaced by its four level-13 children, an equal number of negatives is drawn from the 5% of cells with the lowest mean cosine similarity to the positives in min–max scaled feature space, and a random forest of 1,000 trees on min–max scaled inputs is evaluated by test AUC over 50 stratified 70/30 splits of the augmented labels. R1 draws the negatives at random from cells more than 10 km from any occurrence; R2 keeps the children of one parent on one side of the split; R3 uses one label per occurrence cell; R4 scores the R3 models against realistic background, the test occurrences plus a random 30% of an evaluation

sample of 20,855 cells, the size of the national evaluation pool (prevalence about 0.3%); R5 trains on the occurrence cells and a disjoint training sample of 34,761 cells; R6 replaces the random splits by 100 km blocks; and R7 leaves one province group out. Two controls keep the cosine-dissimilar negatives in the leak-free designs R2 and R3 (R2c, R3c, R4c). From R5 on, RF, XGBoost, bagging-PU and DevNet are evaluated as well. The provincial version of this audit, rebuilt on the 0.1° background of each province, is given in Appendix C.

3.7 Ablation of the deviation loss

DevNet differs from the other learners in three respects: its deviation loss, which treats the training background as unlabelled ground to be described rather than as barren ground to be separated from the occurrences; its rank-normal inputs; and its background-heavy batches. To isolate the first, **DevNet-BCE** keeps DevNet's network, inputs, batches, training length and five-restart ensemble but replaces the deviation loss by binary cross-entropy with the background labelled 0. DevNet-BCE is not a competing prospectivity method but DevNet with one component changed. The tree-based pair, RF (background labelled 0) and bagging-PU (background unlabelled), gives the same contrast for tree ensembles.

The premise of treating the background as unlabelled is that it contains undiscovered occurrences and ground that resembles them. If the premise matters, a loss that treats the background as barren should suffer more when the background holds such ground, and a semi-supervised loss should profit more from a large unlabelled pool and when labels are scarcest. We tested these three predictions, written down before the runs, with 20 repeats instead of six, with and without the point-level map classes, under random cross-validation and 100 km blocks, first in Superior and then, unchanged, in Cordillera and Grenville. *(E1) Mislabelled ground near occurrences.* The main design excludes background within 10 km of every occurrence. Here 0, 20 or 50 additional background points per occurrence, drawn uniformly in a 2–10 km annulus around each occurrence, were added to the training background; under random cross-validation this includes ground around the held-out occurrences, as in practice the unlabelled pool contains ground around undiscovered deposits. Prediction: DevNet-BCE loses more AP than DevNet, and RF more than bagging-PU. *(E2) Size of the unlabelled pool.* The training background was a nested random subset of 625, 1,250, 2,500 or 5,000 points. Prediction: DevNet's lead over DevNet-BCE grows with the pool. *(E3) Label scarcity.* All but $k = 1, 2, 3, 5$ or 8 training positives per fold were dropped, the same ones for every model. Prediction: DevNet is ahead of DevNet-BCE for $k \leq 3$, more so without map classes. For each pair of models and each repeat (identical folds for both models and all conditions) we computed the predicted effect (E1: the difference between the two models' AP changes from 0 to 50 added points; E2: the change in their AP gap from 625 to 5,000 points; E3: their AP difference averaged over $k = 1$–3) and report its median and the number of the 20 repeats in which it favours the unlabelled-background model.

3.8 Other analyses

Nested design selection. DevNet's input transformation and batch composition were settled with diagnostic runs on these three provinces, so its scores may carry selection optimism. To measure it, the choice among the four combinations of z-scored or quantile-normal inputs and 16 or 240 background rows per step was made inside every outer training fold by the inner spatial cross-validation described above (two restarts per candidate), and the chosen design, refitted on the whole outer training fold, scored the outer test fold.

Map stability. Each Superior occurrence was left out in turn and the production models were refitted on the other 14; for every grid node we report the share of these 15 models that place it in the province's top 5%. Prediction–area curves (Yousefi & Carranza, 2015) were computed from the out-of-fold ranks, with the evaluation background as a proxy for area.

Province diagnostics. To describe, independently of any model, why skill differs between provinces, each label set was characterised by its number of independent sites (occurrences merged by single linkage at 10 km), the share of occurrences that share a site with another and the median distance to the nearest other occurrence. The signal shared by distant occurrences was measured with a far-neighbour index: with features expressed as percentile ranks, it is, for each occurrence, the share of background points more than 100 km away that are closer to it in feature space than the most similar occurrence more than 100 km away. A value near 0 means that distant occurrences resemble one another more than they resemble almost any background. The median over occurrences, computed on the geophysical and structural layers, is compared with the same statistic for 200 sets of background points drawn at random to pose as occurrences.

3.9 Synthetic positives

To test whether synthetic positives can compensate for label scarcity, each training fold's real positives were augmented with $m \times n_{\text{pos}}$ synthetic positives ($m = 1, 3, 10$) before RF and DevNet were retrained; the held-out positives were always real. Three generators were compared, all fitted in the standardised feature space of the training fold only: Gaussian jitter of real positives ($\sigma = 0.1$ standard deviations); SMOTE-style interpolation between a positive and one of its three nearest positive neighbours (Chawla, Bowyer, Hall, & Kegelmeyer, 2002); and a tabular Wasserstein GAN with gradient penalty (WGAN-GP; Arjovsky, Chintala, & Bottou, 2017; Goodfellow et al., 2014; Gulrajani, Ahmed, Arjovsky, Dumoulin, & Courville, 2017) with an eight-dimensional latent input, two hidden layers of 64 units in generator and critic, five critic updates per generator update, gradient-penalty weight 10 and 2,000 generator updates, trained on the fold's 12 positives. Synthetic geology columns were snapped back to valid one-hot and 0/1 values. To see what the generators produce, each was also fitted to all 15 Superior positives and 150 samples were characterised by the distance to the nearest real positive divided by the median nearest-neighbour distance among real positives (a value near 0 indicates copying), by the share of samples whose

nearest labelled neighbour is a positive rather than a background point, and by their DevNet score percentile.

3.10 Interpretation and mapping

Production models were fitted to all positives and the training background of each region. SHAP values (Lundberg & Lee, 2017) were computed with TreeSHAP for RF (Lundberg et al., 2020) and DeepSHAP for DevNet on an explain set of all positives plus evaluation-background points (200 in total), with 100 training-background points as DevNet's reference sample; DevNet's SHAP values refer to its quantile-transformed inputs, which leaves the rank-based directions defined below unaffected. DevNet SHAP values were averaged over the five ensemble members. Importance is mean |SHAP|. The *direction* of a feature's effect is the Spearman correlation between the feature's value and its SHAP value over the explain set (for one-hot dummies, the sign of the mean SHAP difference between presence and absence), with $|\rho| < 0.1$ reported as no clear direction. We do not use the sign of the mean SHAP value, which reflects where the explained points lie relative to the explainer's baseline rather than whether higher feature values raise the score: for RF with balanced class weights the TreeSHAP baseline is near 0.5 while nearly all explained points score near 0, so that rule would label every RF feature as "higher → less prospective". Prospectivity maps classify each model's scores on a 0.1° prediction grid into the top 1%, 5% and 10%, because DevNet's anomaly score and RF's probability are not on a common scale. As an external check, the production models scored Clay-Howells, a drill-tested Fe–REE–Nb prospect with reported resources on the KSZ (Ontario Geological Survey, 2011) that is not among the training labels, at its recorded location.

4 Results

4.1 National scale

4.1.1 The national S2-grid design, corrected rung by rung

Rebuilt on our national data, the design of Parsa et al. (2024) gives a test AUC of 1.000 (Table 2; Fig. 5), as high as the published 0.91–0.96 or higher. In 97% of the test positives at least one sibling child cell of the same occurrence is in the training set. Two elements of the design produce this value, and each suffices on its own. With the negatives drawn at random (R1) the AUC stays at 0.996, because the siblings remain; with the siblings kept together (R2) it falls to 0.920, and with one label per occurrence cell (R3) to 0.880. Conversely, cosine-dissimilar negatives without any sibling leak (R2c, R3c) give 1.000 and 0.997. The ground least similar to the occurrences is easy to reject, and a model trained against it is worse at the real task: scored against realistic background, the R3c models reach an AP of 0.091, the R3 models trained against random negatives 0.184.

Up to R3 the test set is balanced, so AP and AUC nearly coincide. At a realistic prevalence of about 0.3% (R4) the AUC of the R3 models is unchanged (0.893), but their AP is 0.184: most cells ranked above a held-out occurrence are background. Training on the shared background changes little (R5; AP 0.292 for the 1,000-tree

Table 2 The national S2-grid design of Parsa et al. (2024) rebuilt on the national data and corrected one element at a time: random forest of 1,000 trees on min–max inputs, mean over 50 stratified 70/30 splits (R0–R5), six repeats of 100 km blocks (R6) or six seeds per held-out province group (R7). Test set: balanced (50% positives) or realistic background (about 0.3% positives). Sibling: share of test positives with a sibling child cell in the training set. R2c–R4c: controls that keep the cosine-dissimilar negatives in the leak-free designs.

Rung	Change	Test set	AUC	AP	Sibling
R0	Design of Parsa et al.: four L13 children per occurrence cell; negatives from the 5% least cosine-similar cells; child-level random 70/30	balanced	1.000	1.000	97%
R1	Negatives at random (>10 km from occurrences)	balanced	0.996	0.997	97%
R2	Children of one parent cell kept on one side	balanced	0.920	0.924	
R2c	control: R2 with cosine-dissimilar negatives	balanced	1.000	1.000	
R3	No augmentation: one label per occurrence cell	balanced	0.880	0.892	
R3c	control: R3 with cosine-dissimilar negatives	balanced	0.997	0.997	
R4	R3 models scored against realistic background	0.3%	0.893	0.184	
R4c	control: R3c models against realistic background	0.3%	0.859	0.091	
R5	Trained on the occurrence cells and a shared background sample	0.3%	0.903	0.292	
R6	100 km spatial blocks, background purged	0.3%	0.840	0.034	
R7	Leave one province group out: Superior held out	0.5%	0.631	0.007	
	Cordillera held out	0.9%	0.745	0.246	
	Grenville held out	0.4%	0.570	0.025	
	rest of Canada held out	0.1%	0.713	0.135	

Table 3 Models from R5 on: AP (AUC) under random 70/30 splits with the shared background (R5), 100 km blocks (R6) and with Superior, Cordillera or Grenville held out (R7).

Model	R5 random	R6 100 km	R7 Superior	R7 Cordillera	R7 Grenville
RF (Parsa et al.)	0.292 (0.90)	0.034 (0.84)	0.007 (0.63)	0.246 (0.74)	0.025 (0.57)
RF	0.281 (0.86)	0.030 (0.76)	0.005 (0.60)	0.169 (0.75)	0.018 (0.59)
XGBoost	0.230 (0.92)	0.021 (0.86)	0.007 (0.72)	0.046 (0.74)	0.026 (0.65)
Bagging-PU	0.254 (0.91)	0.037 (0.88)	0.013 (0.85)	0.028 (0.72)	0.014 (0.59)
DevNet	0.044 (0.86)	0.016 (0.78)	0.015 (0.68)	0.017 (0.56)	0.006 (0.32)

forest). Spatial blocks remove the remaining optimism (R6): with 100 km blocks the AP is 0.034 and the AUC 0.840. Leaving out a whole province (R7), the AUC falls to 0.570–0.745 and the AP for Superior to 0.007, about the random-ranking value of 0.005. From R5 on, RF, XGBoost, bagging-PU and DevNet behave alike (Table 3): the tree ensembles reach an AP of 0.23–0.29 under random splits and DevNet 0.044, and all models 0.016–0.037 under 100 km blocks.

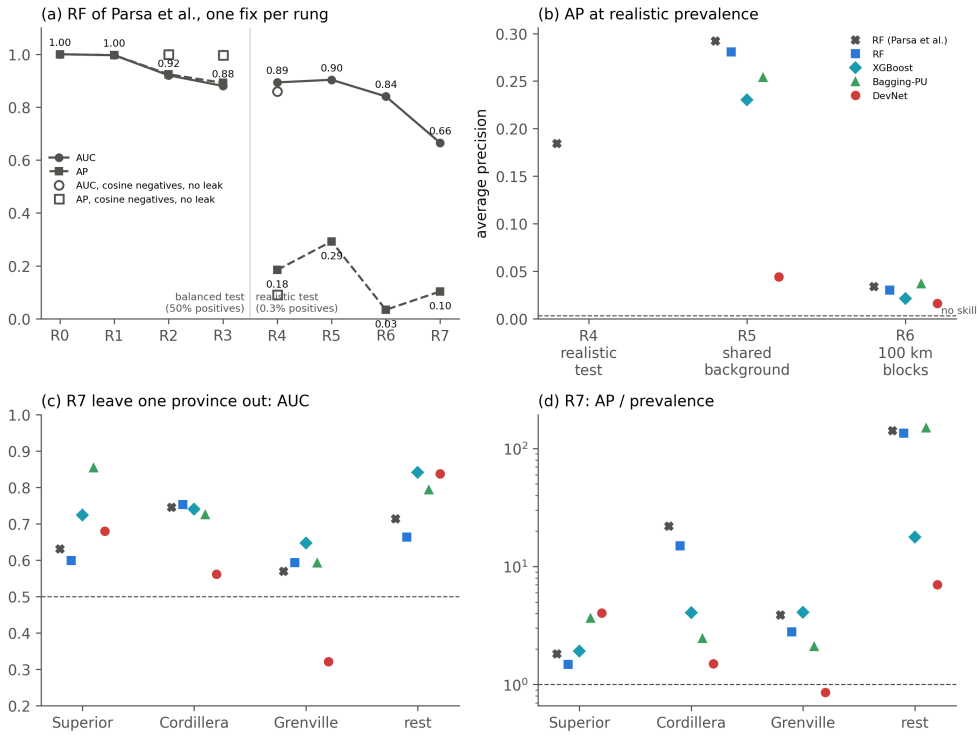


Fig. 5 The national S2-grid design corrected rung by rung. (a) Random forest of Parsa et al. (2024): AUC (circles) and AP (squares); open symbols keep the cosine-dissimilar negatives without the sibling leak; left of the vertical line the test set is balanced, right of it realistic. R7: mean over the four held-out province groups. (b) AP of five models at realistic prevalence; dashed: random ranking. (c, d) Leave one province out: AUC and AP divided by prevalence (log scale).

4.1.2 National benchmark

Under random cross-validation on the national table, RF and bagging-PU reach an AP of 0.308 and 0.302 at a prevalence of 0.32% (Table 4). Much of this is spatial: the best AP is 0.125 with 50 km blocks, 0.061 with 100 km and 0.021 with 200 km blocks. DevNet is the weakest of the evidence-based models (AP 0.040 under random splits and 0.020 under 100 km blocks), well below its own cross-entropy ablation DevNet-BCE (0.131 and 0.039), and nnPU fails (0.008–0.009). AUC hides most of this: for every model but nnPU it stays between 0.76 and 0.94 under every protocol.

On the points of Superior alone, the national models rank the province's occurrences worse than the provincial models under 100 km blocks, for every model (Table 5): DevNet 0.207 against 0.343, RF 0.109 against 0.168 and XGBoost 0.080 against 0.123. Because the inputs take identical values at these points, the difference is the national model itself, with its inputs pruned on the national table and its national training data.

Table 4 National benchmark: AP (AUC) of the pooled out-of-fold scores, median over six repeats; 66 occurrences against 20,855 evaluation-background points (random ranking: AP 0.003). Bold: highest AP in each row.

Split	DevNet	DevNet-BCE	RF	XGBoost	Bagging-PU	nnPU
random	0.040 (0.88)	0.131 (0.89)	**0.308** (0.93)	0.227 (0.94)	0.302 (0.94)	0.009 (0.65)
50 km	0.027 (0.85)	0.042 (0.86)	0.105 (0.85)	0.075 (0.89)	**0.125** (0.91)	0.008 (0.65)
100 km	0.020 (0.80)	0.039 (0.83)	0.051 (0.83)	0.050 (0.88)	**0.061** (0.89)	0.008 (0.64)
200 km	0.014 (0.76)	0.019 (0.78)	**0.021** (0.78)	0.015 (0.78)	0.017 (0.82)	0.008 (0.63)

Table 5 The same Superior points ranked by the provincial and by the national model: AP (median over six repeats) under random cross-validation and 100 km blocks. Inputs take identical values in both.

	Random CV		100 km blocks	
Model	Provincial	National	Provincial	National
DevNet	0.445	0.243	0.343	0.207
RF	0.271	0.310	0.168	0.109
XGBoost	0.275	0.188	0.123	0.080
Bagging-PU	0.219	0.201	0.129	0.070
nnPU	0.152	0.048	0.157	0.052

4.1.3 Transfer between provinces

A province's model does not predict another province (Fig. 6). Within each province, random cross-validation gives an AP of 0.13–0.45 (nnPU excluded); applied to another province, the tree ensembles reach 0.004–0.012, about the random-ranking values of 0.004–0.009, and in 25 of the 36 model–province pairs the AUC is below 0.5: the model ranks the other province's occurrences below its background. The one exception is the neural models trained in Cordillera and applied to Superior (AUC 0.79 for DevNet and 0.85 for DevNet-BCE; AP 0.038 and 0.073 against 0.005). Leaving a province out of a national model gives the same picture (Table 6): held-out Grenville is ranked at or below chance by every model (AUC 0.30–0.68, AP $\leq$ 0.026), held-out Superior reaches an AP of at most 0.051 and held-out Cordillera of at most 0.159 (RF).

The provinces are distinct domains in the inputs. A random forest separates the training background of any two province groups with an AUC of 0.999–1.000, and the occurrences of Superior, Cordillera and Grenville from one another with 0.99–1.00. Several inputs mark occurrences in opposite directions in different provinces. Taken alone, LAB depth separates Superior occurrences from their background with an AUC of 0.22 (shallower LAB at occurrences) but Cordilleran ones with 0.67 (deeper), Moho depth with 0.35 and 0.78, and distance to regional greenstone with 0.23 and 0.63; deep-source magnetics marks Cordilleran and Grenville occurrences in opposite directions (0.25 and 0.81).

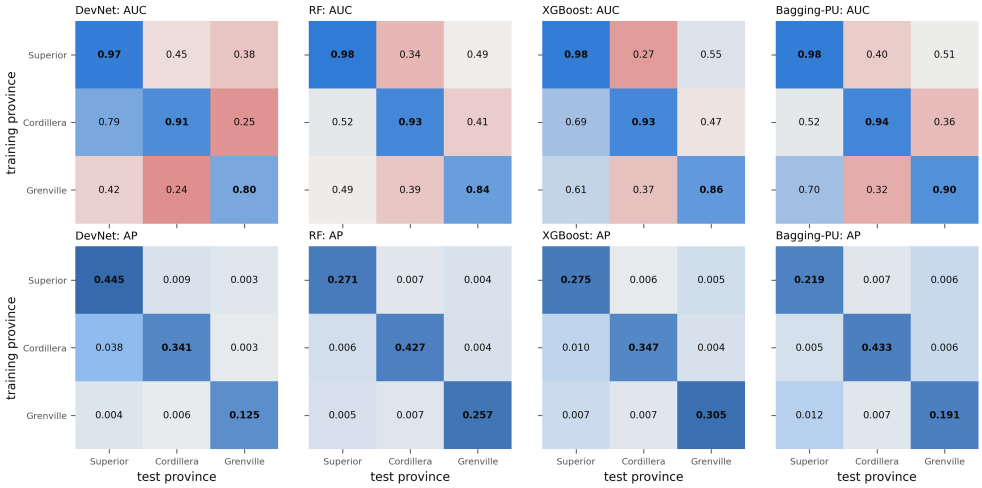


Fig. 6 Transfer between provinces. Top: AUC; bottom: AP, coloured by AP divided by the test province's prevalence (log scale). Rows: province whose model (inputs, capacity, occurrences, training background) is applied; columns: province whose occurrences and evaluation background are scored (median of six seeds). The diagonal is random cross-validation within the province (Table 9); leave-one-province-out results are in Table 6.

Table 6 Leave one province out: AP (AUC) for the held-out province group's occurrences and evaluation background, median of six seeds, national models trained without the group. No skill: AP of a random ranking.

Model	Superior	Cordillera	Grenville	Rest of Canada
DevNet	0.051 (0.71)	0.011 (0.57)	0.003 (0.32)	0.014 (0.80)
DevNet-BCE	0.038 (0.76)	0.011 (0.56)	0.003 (0.30)	0.045 (0.77)
RF	0.016 (0.68)	0.159 (0.80)	0.014 (0.61)	0.177 (0.77)
XGBoost	0.013 (0.78)	0.045 (0.74)	0.026 (0.68)	0.106 (0.82)
Bagging-PU	0.022 (0.89)	0.016 (0.66)	0.011 (0.59)	0.191 (0.81)
nnPU	0.026 (0.72)	0.011 (0.50)	0.004 (0.46)	0.003 (0.72)
No skill	0.0050	0.0089	0.0040	0.0009

4.1.4 Does pooling the country help a province?

Adding the rest of Canada to a province's training data does not improve its predictions (Table 7; Appendix B, Fig. B2). Under 100 km blocks, the national training background outside the province lowered the AP in 11 of 15 model–province combinations, and under random cross-validation in 14 of 15; DevNet lost most (Superior 0.343 to 0.123). The national occurrences outside the province raised the AP in 10 of 15 combinations under blocks, mostly for the tree ensembles and in Grenville, where every AP stays close to the random-ranking value. Adding both, a national model in the province's own specification, lowered the AP in 11 of 15. In Superior the best

Table 7 Pooling the country into a province's training data: AP (median over six repeats) under 100 km blocks for the province's model trained on its own data, plus the national training background outside it, plus the national occurrences outside it, or plus both; change from the province's own data in brackets. Same held-out occurrences, evaluation background and folds in every column.

Province	Model	Own data	+ background	+ occurrences	+ both
Superior	DevNet	0.343	0.123 (-0.220)	0.240 (-0.103)	0.206 (-0.137)
	DevNet-BCE	0.346	0.210 (-0.136)	0.242 (-0.104)	0.234 (-0.112)
	RF	0.168	0.075 (-0.094)	0.129 (-0.039)	0.103 (-0.065)
	XGBoost	0.123	0.091 (-0.032)	0.115 (-0.008)	0.051 (-0.072)
	Bagging-PU	0.129	0.139 (+0.010)	0.130 (+0.001)	0.069 (-0.059)
Cordillera	DevNet	0.055	0.020 (-0.035)	0.072 (+0.017)	0.023 (-0.032)
	DevNet-BCE	0.054	0.032 (-0.022)	0.050 (-0.004)	0.031 (-0.023)
	RF	0.046	0.043 (-0.003)	0.051 (+0.005)	0.035 (-0.011)
	XGBoost	0.040	0.042 (+0.003)	0.049 (+0.009)	0.038 (-0.001)
	Bagging-PU	0.049	0.054 (+0.005)	0.050 (+0.000)	0.065 (+0.016)
Grenville	DevNet	0.007	0.003 (-0.004)	0.013 (+0.006)	0.003 (-0.004)
	DevNet-BCE	0.007	0.003 (-0.004)	0.022 (+0.015)	0.003 (-0.004)
	RF	0.008	0.008 (-0.000)	0.015 (+0.007)	0.015 (+0.007)
	XGBoost	0.006	0.007 (+0.001)	0.019 (+0.013)	0.016 (+0.010)
	Bagging-PU	0.009	0.005 (-0.004)	0.014 (+0.005)	0.012 (+0.002)

model remains one trained on Superior alone (under 100 km blocks DevNet-BCE 0.346 and DevNet 0.343; best pooled 0.242).

4.1.5 Sample size

Under national random cross-validation the AP of the tree ensembles rises steadily with the number of training occurrences and has not levelled off at 52 (RF 0.046 with three, 0.067 with ten and 0.308 with all; Appendix B, Fig. B3); with ten or fewer training occurrences no model exceeds an AP of 0.07. DevNet stays between 0.019 and 0.042 at every count.

4.1.6 Why DevNet is weak nationally

Training is not the problem: scaling DevNet's steps with the size of the background (about seven times more) gives an AP of 0.054 under random splits and 0.023 under 100 km blocks, against 0.040 and 0.020. Of the three predictions (Table 8; Fig. 7), P1 fails: ranking the scores within each province group does not raise DevNet's AP (0.034), and it lowers RF's, so the weakness is not a common offset between provinces. DevNet does, however, place 70% of its national top 10% of background in Grenville and Cordillera, against 53% for RF, although these provinces hold 19% of the evaluation background. P2 holds: with the background of its own province as reference, DevNet's AP doubles (0.085 and 0.044), a larger relative gain than RF's (0.395 and 0.080, factors of 1.3 and 1.6). P3 fails: province one-hot inputs leave DevNet unchanged (0.035 and 0.020) while they help DevNet-BCE (0.190 and 0.057). Even with a local

Table 8 Why DevNet is weak nationally: national AP under random cross-validation / 100 km blocks as in the benchmark, with one model per province group trained against its own background (local reference, P2), and with the province group as one-hot inputs (P3). Within provinces: random-CV AP after ranking the scores within each province group (P1). Top 10%: share of the national top 10% of the evaluation background that lies in Grenville or Cordillera (19% of the evaluation background).

Model	As benchmarked	Local reference	Province one-hot	Within provinces	Top 10%
DevNet	0.040 / 0.020	0.085 / 0.044	0.035 / 0.020	0.034	70%
DevNet-BCE	0.131 / 0.039	0.142 / 0.060	0.190 / 0.057	0.091	68%
RF	0.308 / 0.051	0.395 / 0.080	0.298 / 0.049	0.214	53%

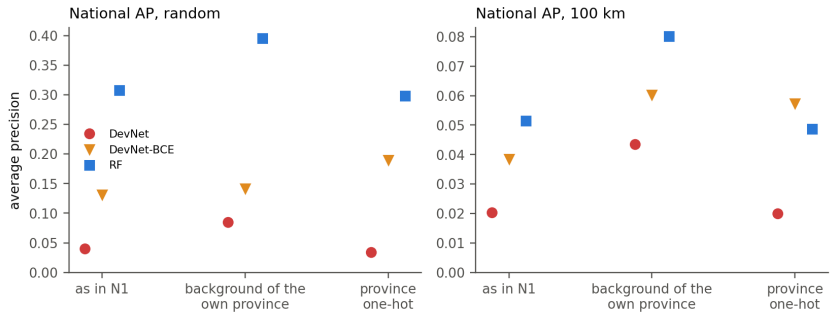


Fig. 7 National AP of DevNet, DevNet-BCE and RF as benchmarked, with one model per province group trained against its own background, and with province one-hot inputs, under (a) random cross-validation and (b) 100 km blocks.

reference, DevNet stays below DevNet-BCE (0.142 and 0.060), so the deviation loss itself, not only the mixed background, is at a disadvantage at the national scale.

4.2 Provinces

The rest of the results concern the three provinces modelled separately, with their own inputs and capacity, as in the main analysis; XGBoost is added to the models.

4.2.1 Model comparison under random cross-validation

In Superior all models except nnPU rank held-out occurrences far above chance (Table 9): AP is 0.22–0.45 against a random-ranking value of 0.005, and AUC is 0.97–0.98. DevNet has the highest median AP (0.445) and RF the highest AUC (0.983). DevNet's AP exceeds RF's, XGBoost's (0.275), bagging-PU's and nnPU's in all six repeated splits, by 0.18–0.26 on repeat-averaged ranks (Table 10). nnPU, which shares DevNet's network and inputs, reaches an AP of only 0.152 and an AUC of 0.838.

The contrasting regions behave differently. In Cordillera, bagging-PU and RF lead (AP 0.433 and 0.427), and DevNet (0.341; lower than RF in all six repeats) and XGBoost (0.347) trail. In Grenville all models are weak; XGBoost leads (0.305), followed by RF (0.257), both ahead of DevNet (0.125) in all six repeats. nnPU fails

Table 9 Pooled out-of-fold AUC and AP (median and 95% interval over six repeated 5-fold random splits). Bold: best of the five models in each region.

Region (n_{pos})	Model	AUC	AP
Superior (15)	DevNet	0.972	**0.445** [0.404, 0.474]
	RF	**0.983**	0.271 [0.165, 0.289]
	XGBoost	0.976	0.275 [0.257, 0.353]
	Bagging-PU	0.976	0.219 [0.158, 0.264]
	nnPU	0.838	0.152 [0.125, 0.191]
Cordillera (27)	DevNet	0.914	0.341 [0.299, 0.394]
	RF	0.931	0.427 [0.368, 0.516]
	XGBoost	0.927	0.347 [0.316, 0.380]
	Bagging-PU	**0.936**	**0.433** [0.304, 0.507]
	nnPU	0.553	0.022 [0.019, 0.029]
Grenville (12)	DevNet	0.799	0.125 [0.054, 0.171]
	RF	0.838	0.257 [0.182, 0.489]
	XGBoost	0.857	**0.305** [0.207, 0.494]
	Bagging-PU	**0.898**	0.191 [0.172, 0.255]
	nnPU	0.542	0.008 [0.006, 0.009]

Table 10 Paired comparisons of DevNet with each other model (random cross-validation): difference in AP and AUC computed on repeat-averaged out-of-fold ranks, with 95% stratified bootstrap interval, and the number of the six repeats in which DevNet has the higher AP.

Region	Comparison	ΔAP [95%]	AP wins	ΔAUC [95%]
Superior	DevNet − RF	+0.210 [+0.001, +0.424]	6/6	-0.012 [-0.032, +0.004]
Superior	DevNet − XGBoost	+0.182 [-0.033, +0.382]	6/6	-0.007 [-0.030, +0.012]
Superior	DevNet − Bagging-PU	+0.256 [+0.026, +0.471]	6/6	-0.006 [-0.028, +0.012]
Superior	DevNet − nnPU	+0.244 [+0.053, +0.448]	6/6	+0.111 [+0.037, +0.189]
Cordillera	DevNet − RF	-0.093 [-0.228, +0.092]	0/6	-0.032 [-0.098, +0.012]
Cordillera	DevNet − XGBoost	-0.059 [-0.191, +0.128]	3/6	-0.021 [-0.073, +0.015]
Cordillera	DevNet − Bagging-PU	-0.084 [-0.222, +0.103]	1/6	-0.021 [-0.080, +0.023]
Cordillera	DevNet − nnPU	+0.353 [+0.189, +0.570]	6/6	+0.359 [+0.199, +0.517]
Grenville	DevNet − RF	-0.258 [-0.446, -0.076]	0/6	-0.050 [-0.147, +0.077]
Grenville	DevNet − XGBoost	-0.246 [-0.438, -0.054]	0/6	-0.075 [-0.153, -0.007]
Grenville	DevNet − Bagging-PU	-0.162 [-0.305, -0.018]	0/6	-0.108 [-0.180, -0.044]
Grenville	DevNet − nnPU	+0.194 [+0.025, +0.453]	6/6	+0.262 [+0.044, +0.467]

in both provinces (AUC 0.54–0.55). That Superior is the best-predicted of the three provinces agrees with the province-level success rates of Parsa et al. (2024).

4.2.2 Spatially blocked validation

Holding out whole blocks widens the gap between DevNet and the tree-based and PU learners in Superior (Fig. 8; Table 11). DevNet's AP is 0.418, 0.343 and 0.252 with 50, 100 and 200 km blocks. RF's falls from 0.271 under random splits to 0.192, 0.168 and 0.151, XGBoost's from 0.275 to 0.202, 0.123 and 0.131, and bagging-PU's to 0.148, 0.129 and 0.090. DevNet has the higher AP than RF, bagging-PU and nnPU in all six repeats at every block size, and than XGBoost in 17 of the 18 (Table 12), by +0.18 to +0.24 against RF, +0.23 to +0.31 against XGBoost, +0.25 to +0.33 against

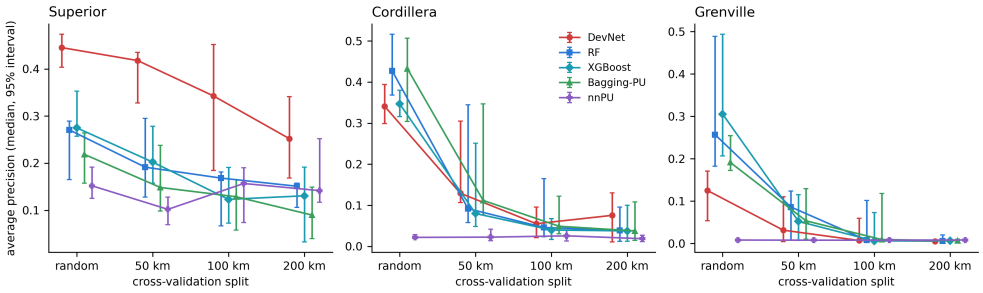


Fig. 8 Average precision under random and spatially blocked cross-validation (median and 95% interval over six repeats). The random-ranking AP is 0.005 (Superior), 0.009 (Cordillera) and 0.004 (Grenville).

bagging-PU and +0.19 to +0.41 against nnPU. With 15 occurrences, however, the 95% intervals of the differences from RF reach down to about zero, so the margin over RF is consistent but not statistically conclusive; the margins over bagging-PU and nnPU are clearer. AUC tells a partly different story with 200 km blocks, where RF (0.880), bagging-PU (0.855) and XGBoost (0.826) exceed DevNet (0.783; Appendix A, Table A1): DevNet ranks most held-out occurrences very high but misses a few almost entirely, notably Kapuskasing and Nemegosenda Lake.

Capture-rate curves (Fig. 9) show where the AP differences arise. Under random splits DevNet and RF capture about half of the held-out occurrences in the top 1% of the ranked pool (53% and 50%), and RF and bagging-PU capture all of them by the top 10% (DevNet 93%). With 50 and 100 km blocks DevNet keeps its top-1% capture (53% and 47%) and captures 90–93% by the top 10%, whereas RF captures 43% in the top 1% and 83–90% by the top 10%, and bagging-PU 30% and 83–90%. With 200 km blocks every model captures 47% or less in the top 1%, and by the top 10% RF and bagging-PU (70%) slightly exceed DevNet (63%). Prediction–area curves (Appendix A, Fig. A1), which summarise the whole ranking rather than its top, do not separate the models.

In Cordillera every model degrades sharply under blocking (AP $\leq$ 0.17 at 50 km and $\leq$ 0.09 at 100–200 km), and in Grenville AP approaches the random-ranking value ($\leq$ 0.009 at 100–200 km). In these regions no difference between models is of practical size: their random-split skill owes much to spatial proximity between training and test occurrences (Section 4.2.6).

4.2.3 Geological map classes

A single map class carries much of the skill in Superior. Removing the alkalic-intrusion column lowers DevNet's AP to 0.345, 0.256, 0.219 and 0.154 under random splits and 50, 100 and 200 km blocks, and RF's to 0.230, 0.126, 0.096 and 0.084: this one class, which covers 6 of the 15 Superior occurrences and 0.5% of the background, accounts for over a third of DevNet's spatially blocked AP. DevNet without it still has a higher AP than RF with it at every block size. Withholding all point-level map classes costs

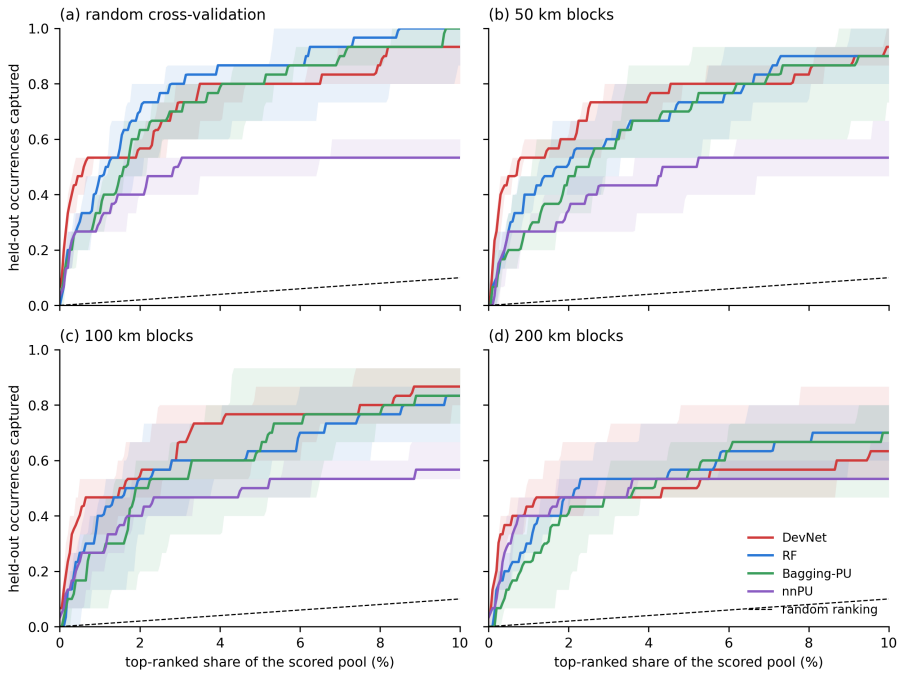


Fig. 9 Superior capture-rate curves: share of held-out occurrences ranked within the top fraction of the scored pool (a proxy for search area) under (a) random cross-validation and (b) 50 km, (c) 100 km and (d) 200 km spatial blocks. Lines: median over six repeats; bands: range. Dashed: random ranking.

DevNet and RF about half or more of their AP: DevNet's falls to 0.183 under random splits and 0.133, 0.109 and 0.079 under blocks, RF's to 0.138 and 0.084, 0.069 and 0.051. Without map classes DevNet still has the highest AP under every block size, narrowly ahead of bagging-PU (0.130, 0.079 and 0.065), while bagging-PU leads under random splits (0.225) and nnPU fails (AP 0.007).

4.2.4 Tuned baselines and design selection

Tuning did not help the tree-based learners in Superior (Table 13). RF-tuned reached an AP of 0.263, 0.163, 0.142 and 0.117 under random splits and 50, 100 and 200 km blocks, and HGB-tuned 0.200, 0.118, 0.087 and 0.063, at or below the default RF; DevNet had the higher AP than both in all six repeats under every protocol. XGBoost, with parameters fixed in advance, behaves like these baselines (0.275, 0.202, 0.123 and 0.131; Table 11). In Cordillera, HGB-tuned has the highest random-split AP (0.470) and in Grenville the default RF (0.257); under blocks all models are close to chance in both provinces.

DevNet's own design selection carries some optimism (Appendix A, Table A2). Choosing the input transformation and batch composition inside each outer training fold gave Superior APs of 0.354, 0.335, 0.312 and 0.190, against 0.445, 0.418, 0.343 and 0.252 for the final configuration, so part of the final configuration's advantage reflects

Table 11 Average precision (median over six repeats) under random and spatially blocked cross-validation. Bold: highest AP in each row. AUC for the same runs is given in Appendix A (Table A1).

Region	Split	DevNet	RF	XGBoost	Bagging-PU	nnPU
Superior	random	**0.445**	0.271	0.275	0.219	0.152
	50 km	**0.418**	0.192	0.202	0.148	0.102
	100 km	**0.343**	0.168	0.123	0.129	0.157
	200 km	**0.252**	0.151	0.131	0.090	0.142
Cordillera	random	0.341	0.427	0.347	**0.433**	0.022
	50 km	**0.129**	0.092	0.080	0.112	0.023
	100 km	**0.055**	0.046	0.040	0.049	0.026
	200 km	**0.076**	0.039	0.038	0.038	0.019
Grenville	random	0.125	0.257	**0.305**	0.191	0.008
	50 km	0.031	**0.086**	0.052	0.053	0.008
	100 km	0.007	0.008	0.006	**0.009**	0.008
	200 km	0.005	0.006	0.006	0.007	**0.008**

Table 12 Paired comparisons of DevNet with each other model under spatially blocked cross-validation, Superior: difference in AP on repeat-averaged ranks with 95% bootstrap interval, and the number of the six repeats in which DevNet has the higher AP.

Blocks	Comparison	ΔAP [95%]	AP wins
50 km	DevNet − RF	+0.240 [-0.004, +0.475]	6/6
50 km	DevNet − XGBoost	+0.302 [+0.062, +0.480]	6/6
50 km	DevNet − Bagging-PU	+0.328 [+0.084, +0.553]	6/6
50 km	DevNet − nnPU	+0.410 [+0.200, +0.577]	6/6
100 km	DevNet − RF	+0.196 [+0.011, +0.368]	6/6
100 km	DevNet − XGBoost	+0.227 [+0.059, +0.411]	6/6
100 km	DevNet − Bagging-PU	+0.245 [+0.014, +0.463]	6/6
100 km	DevNet − nnPU	+0.188 [+0.050, +0.354]	6/6
200 km	DevNet − RF	+0.183 [+0.004, +0.316]	6/6
200 km	DevNet − XGBoost	+0.307 [+0.102, +0.487]	5/6
200 km	DevNet − Bagging-PU	+0.284 [+0.057, +0.478]	6/6
200 km	DevNet − nnPU	+0.216 [+0.040, +0.349]	6/6

having been chosen with these provinces in view. The nested procedure nevertheless still had the higher AP than RF in 23 of 24 repeat–protocol combinations in Superior.

4.2.5 Ablation: what the deviation loss adds

With all labels, replacing DevNet's deviation loss by cross-entropy changes little (Table 14). With the map classes DevNet-BCE has a similar or slightly higher AP than DevNet under every protocol (paired differences −0.056 to −0.013; DevNet higher in at most three of six repeats). Without the map classes DevNet is ahead in 23 of the 24 repeat–protocol combinations, by up to 0.05. DevNet's lead over the tree-based and PU learners therefore comes mainly from its other two components, rank-normal inputs and background-heavy batches (Appendix F): DevNet-BCE, which keeps them, is ahead of RF, bagging-PU and nnPU by similar margins.

Table 13 Tuned tree-based baselines: AP (median over six repeats). RF and HGB (tuned): hyperparameters chosen inside each outer training fold by inner 100 km spatial cross-validation. Bold: highest AP in each row.

Region	Split	DevNet	RF	RF (tuned)	HGB (tuned)
Superior	random	**0.445**	0.271	0.263	0.200
	50 km	**0.418**	0.192	0.163	0.118
	100 km	**0.343**	0.168	0.142	0.087
	200 km	**0.252**	0.151	0.117	0.063
Cordillera	random	0.341	0.427	0.435	**0.470**
	50 km	**0.129**	0.092	0.110	0.101
	100 km	**0.055**	0.046	0.048	0.036
	200 km	**0.076**	0.039	0.036	0.037
Grenville	random	0.125	**0.257**	0.164	0.203
	50 km	0.031	0.086	0.067	**0.097**
	100 km	0.007	0.008	**0.009**	0.006
	200 km	0.005	**0.006**	0.005	0.006

The three tests of the semi-supervised premise (Table 15; Fig. 10) give a clear but conditional answer. Labelling ground near the occurrences as barren (E1) hurt the model that treats the background as barren more than its unlabelled-background counterpart in most comparisons. In Superior, adding 50 near-occurrence points per occurrence lowered RF's AP from 0.259 to 0.136 but bagging-PU's only from 0.211 to 0.193 (bagging-PU lost less in 18 of 20 repeats); in Cordillera DevNet-BCE lost more than DevNet in 17 of 20 repeats, and in Grenville both DevNet-BCE and RF lost more than their counterparts (17 and 18 of 20). Over the three provinces, with and without map classes, the unlabelled-background model of each pair lost less in five of six comparisons and clearly more in none. With one to three training positives (E3), DevNet was ahead of DevNet-BCE without map classes in Superior at every label count (19 of 20 repeats on average over $k = 1$–3, and 15–20 of 20 at each k) and in Grenville (17 of 20); with the map classes the two losses were level. The size of the unlabelled pool (E2) made no consistent difference, contrary to our prediction. Under 100 km blocks in Cordillera and Grenville, where every model is close to chance, all effects are near zero.

4.2.6 How much of the score is location?

A random forest given only coordinates reaches an AP of 0.507 in Cordillera under random cross-validation, higher than any evidence-based model (at most 0.470, HGB-tuned), and 0.249 in Grenville, close to RF (0.257) and below only XGBoost (0.305; Table 16). In these two provinces the random-split skill of the tree-based models can therefore be reproduced without any geology: training occurrences cluster, a random split leaves most held-out occurrences close to a training occurrence, and a flexible learner can exploit that proximity, through the coordinates themselves or through spatially smooth layers that act as surrogate coordinates. Under 100 km blocks the coordinate-only model collapses (0.052 and 0.075), as do the evidence-based models (≤ 0.062 and ≤ 0.009).

Table 14 Ablation of the deviation loss, Superior, all labels: AP (median over six repeats) of DevNet and of DevNet-BCE (identical network, inputs, batches and training; binary cross-entropy with the background labelled 0), and the paired difference on repeat-averaged ranks with 95% bootstrap interval and the number of repeats in which DevNet has the higher AP. Without map classes: the lithology, era and contact columns removed.

Inputs	Split	DevNet	DevNet-BCE	ΔAP [95%]	wins
Feature set 3	random	0.445	0.433	-0.034 [-0.086, -0.001]	3/6
	50 km	0.418	0.434	-0.034 [-0.095, +0.004]	0/6
	100 km	0.343	0.346	-0.056 [-0.126, +0.011]	1/6
	200 km	0.252	0.233	-0.013 [-0.062, +0.017]	2/6
Without map classes	random	0.183	0.157	+0.017 [-0.150, +0.237]	6/6
	50 km	0.133	0.094	+0.021 [-0.111, +0.220]	6/6
	100 km	0.109	0.088	-0.010 [-0.135, +0.145]	5/6
	200 km	0.079	0.050	+0.053 [-0.040, +0.227]	6/6

Table 15 Tests of the semi-supervised premise (random cross-validation, 20 repeats). Each entry is the median effect in favour of the model that treats the background as unlabelled (DevNet, bagging-PU), with the number of the 20 repeats in which it is positive. Mislabelled ground: the AP change of the unlabelled-background model minus that of its counterpart when 50 points per occurrence from a 2–10 km annulus are added to the training background. Few labels: the AP difference averaged over one to three training positives per fold.

Test	Pair	Superior	Cordillera	Grenville
Ground near occurrences labelled barren	DevNet vs DevNet-BCE	+0.003 (12/20)	+0.025 (17/20)	+0.044 (17/20)
	Bagging-PU vs RF	+0.089 (18/20)	+0.030 (11/20)	+0.130 (18/20)
One to three training positives	DevNet vs DevNet-BCE	-0.001 (8/20)	+0.004 (16/20)	+0.001 (13/20)
	DevNet vs DevNet-BCE, without map classes	+0.034 (19/20)	+0.003 (12/20)	+0.011 (17/20)

Superior is the exception. Coordinates alone reach only 0.157 under random splits and 0.018 under blocking, far below every evidence-based model except nnPU, so the Superior scores reflect a signature in the evidence rather than location. DevNet uses the evidence better than an L2-regularised logistic regression in every province (random splits: 0.445, 0.341 and 0.125 against 0.374, 0.210 and 0.037; 100 km blocks in Superior: 0.343 against 0.219).

4.2.7 Synthetic positives

The generators behave very differently (Table 18; Fig. 11). Jitter and SMOTE samples lie close to the real positives (median nearest-neighbour ratio 0.10 and 0.24) and almost all have a real positive as their nearest labelled neighbour (100% and 97%): they are near-copies of the training occurrences. WGAN-GP samples are as far from the real positives as the real positives are from one another (ratio 0.91), but for 35% of them the nearest labelled point is a background point: with 15 training points the

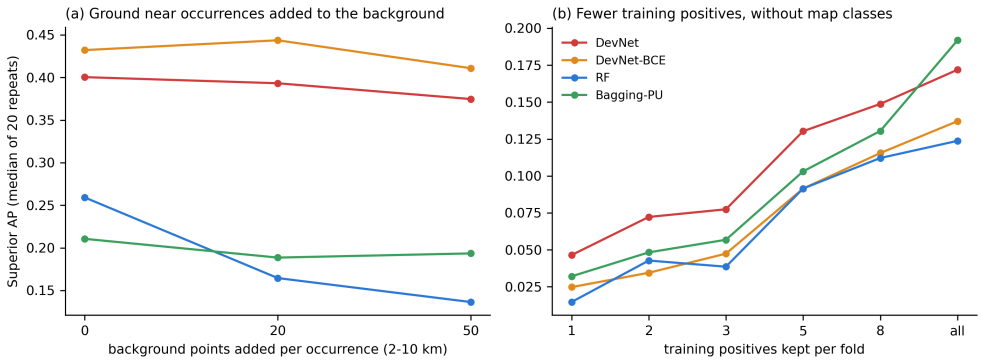


Fig. 10 Tests of the semi-supervised premise, Superior, random cross-validation (median AP over 20 repeats): (a) background points added in a 2–10 km annulus around each occurrence; (b) training positives kept per fold, without the point-level map classes.

Table 16 Geography-only and linear baselines: AP and AUC (median over six repeats) under random cross-validation and 100 km spatial blocks, on identical folds. RF, coordinates only: random forest trained on projected coordinates without any evidence layer.

		Random CV		100 km blocks	
Region	Model	AP	AUC	AP	AUC
Superior	RF, coordinates only	0.157	0.782	0.018	0.750
	RF, all features	0.271	0.983	0.168	0.904
	Logistic regression	0.374	0.924	0.219	0.942
	DevNet	0.445	0.972	0.343	0.950
Cordillera	RF, coordinates only	0.507	0.922	0.052	0.727
	RF, all features	0.427	0.931	0.046	0.776
	Logistic regression	0.210	0.916	0.062	0.826
	DevNet	0.341	0.914	0.055	0.790
Grenville	RF, coordinates only	0.249	0.821	0.075	0.750
	RF, all features	0.257	0.838	0.008	0.538
	Logistic regression	0.037	0.766	0.005	0.624
	DevNet	0.125	0.799	0.007	0.661

generator spreads probability mass into background-like parts of feature space. All three sets of samples receive high DevNet scores (median percentile 0.98).

Because the folds are identical with and without augmentation, the effect is measured as the paired, per-repeat difference in AP (Table 17; Fig. 12). For DevNet no generator helps: the median per-repeat change in AP lies between −0.041 and −0.003, and no configuration raises AP in more than three of the six repeats. For RF, jitter raises AP (median +0.06 to +0.07, in four or five of six repeats), SMOTE lowers it (by 0.070 at $m = 10$), and WGAN-GP changes it little. The best augmented RF (jitter 10×, AP 0.327) remains well below DevNet without augmentation (0.445). Under spatial blocking (Appendix D, Table D8) WGAN-GP at 10× raises RF's AP with 100 km blocks from 0.168 to 0.219, still below unaugmented DevNet (0.343), but with

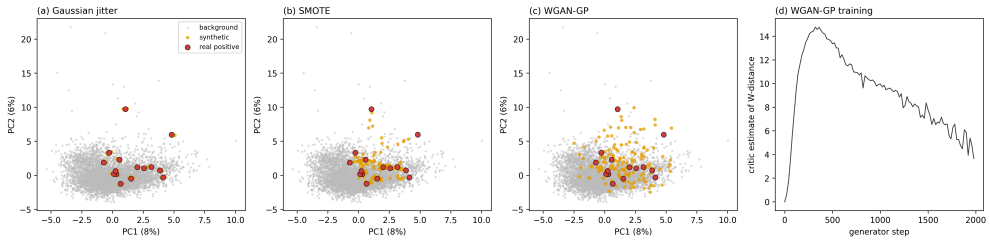


Fig. 11 (a–c) Synthetic positives from each generator (fitted to all 15 Superior positives) in the first two principal components of background and real positives. (d) WGAN-GP critic estimate of the Wasserstein distance during training.

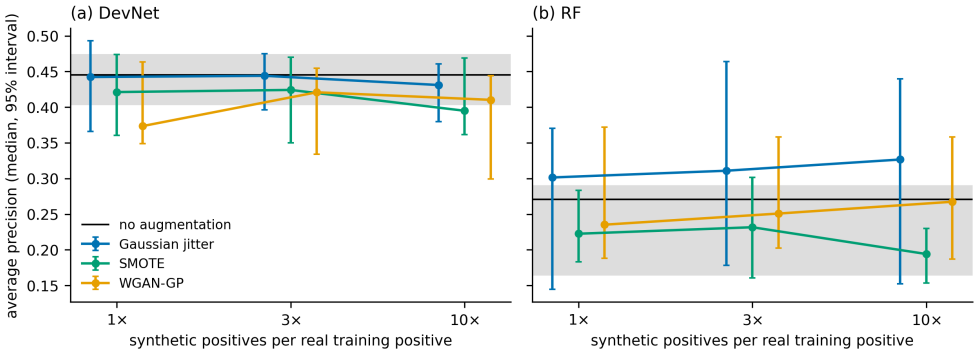


Fig. 12 Superior average precision with fold-safe synthetic positives, by generator and number of synthetic positives per real training positive. Line and grey band: no augmentation (median and 95% interval).

200 km blocks every generator lowers RF's AP or leaves it unchanged. For DevNet the per-repeat changes under blocking are small (medians between −0.031 and +0.001).

4.2.8 Model interpretation

With the rank-based direction rule, RF and DevNet agree on the sign of the effect for all ten of their jointly most important features in each province (Table 19; Appendix E). Read from the sign of the mean SHAP value instead, the two models would appear to disagree on many of these features, an artefact of the two explainers' different baselines (Section 3.10).

The two models agree on directions but weight the evidence differently (Fig. 13). In Superior, DevNet ranks five geological classes of the cell first: extrusive (greenstone) lithology, an age contact within the cell and alkalic intrusive lithology raise the score, and Archean era and tonalite lower it; it then ranks proximity to regional greenstone, Bouguer GLCM correlation (lower values more prospective) and LAB depth. RF ranks the horizontal-gradient magnitude of deep-source magnetics first

Table 17 Effect of fold-safe positive augmentation on held-out performance, Superior (random cross-validation, identical folds; held-out positives always real). ΔAP: paired difference from no augmentation in each of the six repeats (median [min, max]); wins: repeats with higher AP than without augmentation.

Model	Generator	m	AUC	AP	ΔAP per repeat	wins
DevNet	none	–	0.972	0.445 [0.404, 0.474]	–	–
	Gaussian jitter	1×	0.964	0.442 [0.366, 0.493]	-0.003 [-0.093, +0.025]	2/6
	Gaussian jitter	3×	0.973	0.444 [0.396, 0.475]	-0.006 [-0.009, +0.021]	2/6
	Gaussian jitter	10×	0.974	0.431 [0.380, 0.461]	-0.008 [-0.078, +0.004]	1/6
	SMOTE	1×	0.974	0.421 [0.360, 0.474]	-0.025 [-0.045, +0.020]	1/6
	SMOTE	3×	0.972	0.424 [0.350, 0.470]	-0.017 [-0.056, +0.000]	1/6
	SMOTE	10×	0.975	0.395 [0.362, 0.469]	-0.035 [-0.097, +0.013]	2/6
	WGAN-GP	1×	0.968	0.374 [0.349, 0.464]	-0.041 [-0.109, +0.010]	1/6
	WGAN-GP	3×	0.970	0.421 [0.334, 0.454]	-0.016 [-0.123, +0.029]	3/6
	WGAN-GP	10×	0.964	0.410 [0.299, 0.444]	-0.025 [-0.170, -0.005]	0/6
RF	none	–	0.983	0.271 [0.165, 0.289]	–	–
	Gaussian jitter	1×	0.973	0.301 [0.144, 0.371]	+0.059 [-0.065, +0.083]	4/6
	Gaussian jitter	3×	0.983	0.311 [0.178, 0.464]	+0.059 [-0.071, +0.211]	5/6
	Gaussian jitter	10×	0.975	0.327 [0.152, 0.440]	+0.069 [-0.042, +0.159]	4/6
	SMOTE	1×	0.980	0.223 [0.183, 0.283]	-0.021 [-0.066, +0.022]	2/6
	SMOTE	3×	0.984	0.232 [0.161, 0.301]	-0.006 [-0.078, +0.027]	2/6
	SMOTE	10×	0.981	0.194 [0.154, 0.230]	-0.070 [-0.113, +0.002]	1/6
	WGAN-GP	1×	0.963	0.235 [0.188, 0.372]	+0.007 [-0.100, +0.098]	3/6
	WGAN-GP	3×	0.979	0.251 [0.202, 0.358]	+0.018 [-0.066, +0.132]	3/6
	WGAN-GP	10×	0.985	0.267 [0.187, 0.358]	+0.035 [-0.079, +0.115]	4/6

Table 18 What the generators produce (fitted to all 15 Superior positives; 150 samples each). NN ratio: distance to the nearest real positive divided by the median nearest-neighbour distance among real positives (0 = copy). Nearest labelled neighbour: share of samples whose nearest point among positives and training background is a positive. DevNet percentile: median score percentile among the training background under the production DevNet.

Source	NN ratio, median	NN ratio, P10	NN is positive	DevNet pct.
Real occurrences (leave-one-out)	1.00	0.55	–	0.995
Gaussian jitter	0.10	0.08	100%	0.984
SMOTE	0.24	0.06	97%	0.982
WGAN-GP	0.91	0.58	65%	0.985
Background (300 random points)	1.51	–	–	0.500

(lower values more prospective), then LAB depth, proximity to regional greenstone and fault proximity, and gives the geological classes low ranks (20th–34th of 48). Both models score ground as more prospective where the LAB is shallower, close to greenstone belts and faults, where TMI is high, and farther from deep magnetic worms. In Cordillera both models treat Mesozoic terranes and ground far from high-grade metamorphic units as unfavourable and Proterozoic terranes as favourable. In Grenville both rank deep-source magnetics and distance to regional age contacts

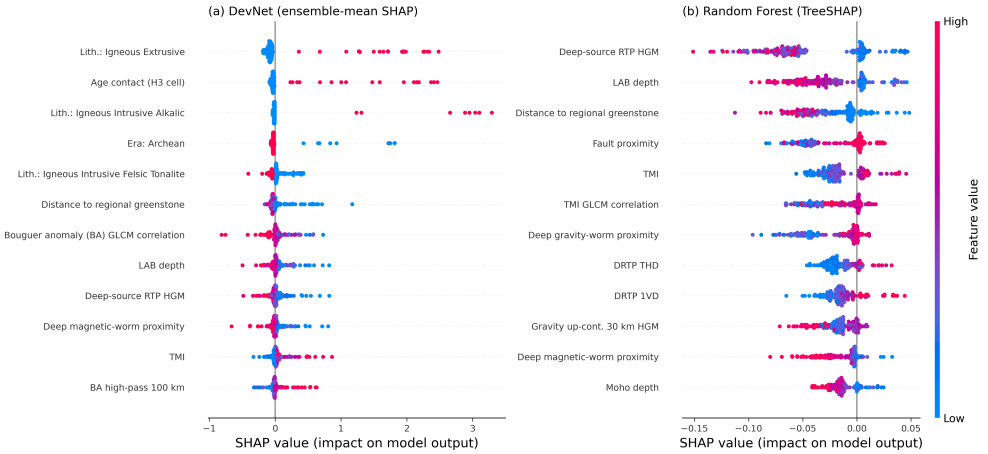


Fig. 13 SHAP summaries for Superior: (a) DevNet (DeepSHAP on its quantile-transformed inputs, mean over the five ensemble members) and (b) RF (TreeSHAP). Each point is one explained sample; colour is the feature value. SHAP magnitudes are not comparable between models (unbounded anomaly score vs probability).

Table 19 Superior: the ten jointly most important features (mean |SHAP|, normalised per model). Rank among 48 features; direction ↑/↓: higher values (or presence) raise/lower the score, with Spearman ρ between feature value and SHAP value.

Feature	DevNet rank	DevNet dir.	RF rank	RF dir.
Deep-source RTP HGM	9	↓ (-0.95)	1	↓ (-0.74)
Lith.: Igneous Extrusive	1	↑ (+0.53)	34	↑ (+0.53)
LAB depth	8	↓ (-0.94)	2	↓ (-0.90)
Distance to regional greenstone	6	↓ (-0.89)	3	↓ (-0.90)
Age contact (H3 cell)	2	↑ (+0.50)	31	↑ (+0.50)
TMI	11	↑ (+0.92)	5	↑ (+0.86)
Lith.: Igneous Intrusive Alkalic	3	↑ (+0.34)	20	↑ (+0.34)
Fault proximity	15	↑ (+0.90)	4	↑ (+0.84)
Bouguer anomaly (BA) GLCM correlation	7	↓ (-0.88)	16	↓ (-0.90)
Deep magnetic-worm proximity	10	↓ (-0.92)	11	↓ (-0.93)

among their four most important features, but given the weak predictive skill there these attributions describe the signal the models attempted to use rather than a reliable control.

4.2.9 Prospectivity maps and external checks

The production DevNet and RF maps (Fig. 14) both concentrate their highest classes in the southern Superior Province, around the eastern-Abitibi cluster and across the

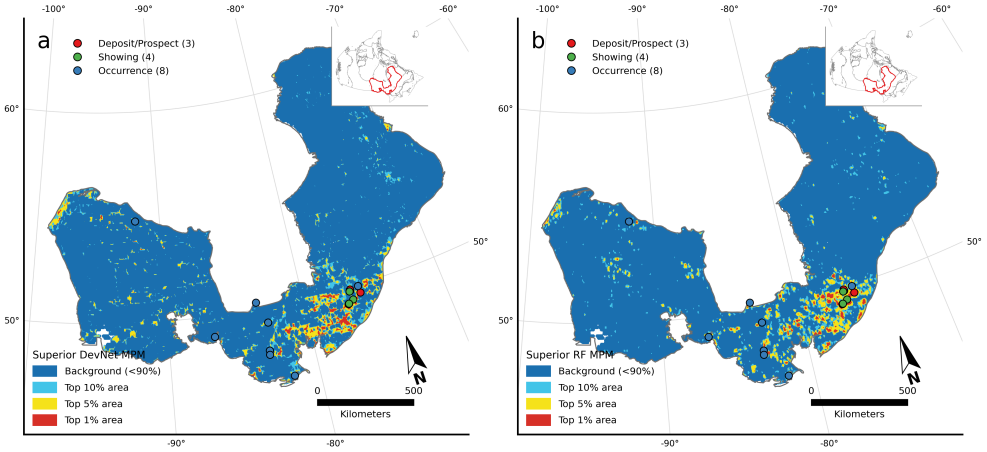


Fig. 14 Superior prospectivity maps from the production (a) DevNet and (b) RF models, classified by score percentile over the mapped area (top 1%, 5% and 10%), with known occurrences by development class (Canada Atlas Lambert). Scores on the 0.1° prediction grid were interpolated to a 2 km raster.

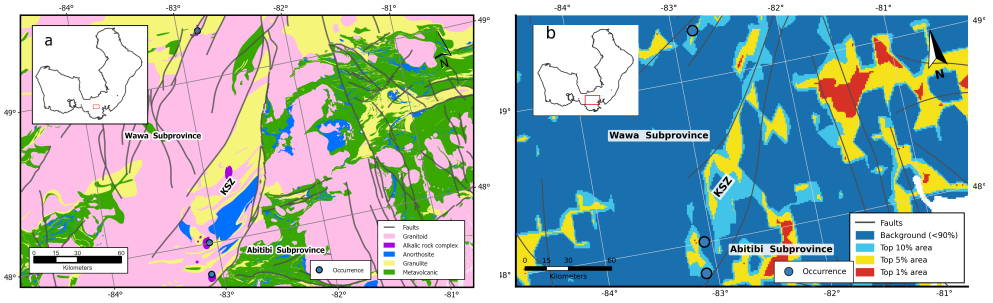


Fig. 15 Kapuskasing area. (a) Regional geology with the Kapuskasing Structural Zone (KSZ) and alkalic rock complexes (Gittins et al., 1967; Percival & West, 1994), compiled from Heather, Percival, Moser, and Bleeker (1995) and Montsion, Thurston, and Ayer (2018). (b) DevNet top 1%, 5% and 10% classes (province-wide percentiles), with faults from the compilation used as an evidence layer, on the same frame as (a).

Abitibi and Wawa subprovinces, but their rankings of the province are only moderately similar (Spearman correlation over the grid 0.37). RF assigns a probability of exactly zero to 83% of the grid, so its classes are drawn from the remaining nodes. In the Kapuskasing area (Fig. 15) DevNet places its top 1% mainly in the Abitibi Subprovince east of the KSZ; along the KSZ itself, the top 5% and 10% classes form discontinuous north-north-east trending patches on its western side. Clay-Howells, the untrained Fe–REE–Nb resource on the KSZ, lies 79 km from the nearest training occurrence; among the province's 8,000 random background points the production models place it at the 96.9th (DevNet) and 99.4th (RF) percentile.

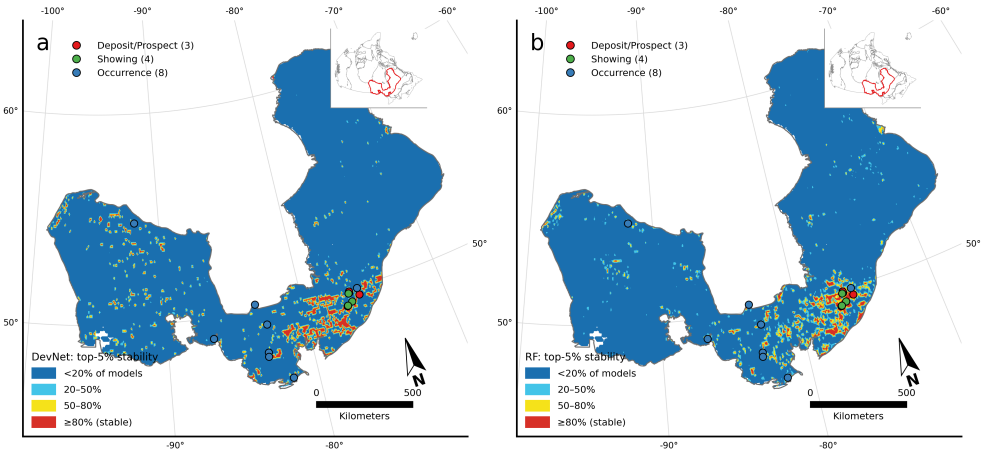


Fig. 16 Stability of the Superior top-5% class with respect to individual occurrences: share of the 15 leave-one-occurrence-out models that place each 2 km cell in the province's top 5%, for (a) DevNet and (b) RF.

The DevNet map is more stable with respect to individual occurrences than the RF map (Fig. 16). When each of the 15 occurrences is left out in turn, 93% of the grid nodes in the full DevNet model's top 5% remain in the top 5% in at least 12 of the 15 refitted models, and 1% in fewer than half of them, whereas for RF the shares are 53% and 28%. RF's top classes therefore depend strongly on which occurrences are known, whereas DevNet's follow properties shared by many of them.

5 Discussion

5.1 Scale: a national model is not a better provincial model

Pooling the occurrences of a country is the obvious response to label scarcity, and national models report high scores (Parsa et al., 2024). Our national results show why such scores do not carry over to prospecting within a province. Under spatial blocks the national models reach an AP of at most 0.06, and a province's model ranks another province's occurrences at or below chance in most pairs. The provinces are not samples of one population: a random forest tells any two of them apart from the background alone, their occurrences are almost as separable, and several inputs, including LAB and Moho depth and distance to greenstone, mark occurrences in opposite directions in Superior and Cordillera. A national model therefore has to learn several mineral-system expressions at once from 66 occurrences, and in the province where most is known it is outperformed by a model trained on that province alone (Superior, 100 km blocks: 0.343 against 0.207 for DevNet). In the terms of Meyer and Pebesma (2021), a province lies outside the area of applicability of a model trained on another, and pooling widens the training domain without making it applicable.

The pooling test separates the two things a national pool adds. More background from other provinces lowered the AP in most model–province combinations, because it changes what the model learns to tell the occurrences from; more occurrences from other provinces helped the tree ensembles somewhat, so the provinces share a little signal, but not enough to outweigh the foreign background. The learning curve explains why a larger pool is still attractive: national AP keeps rising up to the 52 training occurrences available. What would help is more occurrences of the same kind, not more ground of other kinds.

Two methodological points follow. First, an input whose value depends on the extent of the study area, such as a texture quantised over the clipped raster or a map generalised within the study area, is not the same input at two scales. Computing such inputs over the whole country lowered Superior DevNet's blocked AP from 0.343 to 0.291 before any model was changed (Appendix B, Table B4); comparisons between scales are meaningful only once these inputs are computed within fixed units, here the geological provinces. Second, transfer to held-out provinces should be tested before a pooled model is used anywhere its training occurrences do not reach.

5.2 What the semi-supervised formulation adds

Three results bound the answer within a province. First, in Superior DevNet ranks held-out occurrences better than RF, tuned RF, tuned gradient boosting, bagging-PU and nnPU in every repeat of every protocol, and better than XGBoost in all but one, and its lead is largest when the test requires extrapolation: RF and bagging-PU, which learn a sharp decision boundary around the training positives, lose 30–60% of their AP once the nearest training occurrence is 50–200 km away, whereas DevNet keeps most of its AP to 100 km. With 15 positives the margin over RF is not statistically conclusive on its own, and part of DevNet's advantage reflects a design chosen with these provinces in view; but the lead is consistent across all 18 repeat–block combinations, it survives a design search for the tree-based learners, and DevNet chosen by nested selection still beats RF in 23 of 24 combinations.

Second, the ablation shows where most of this lead comes from. With all labels and the map classes, DevNet-BCE, which keeps DevNet's network, rank-normal inputs and background-heavy batches but labels the background barren, does as well as DevNet. Most of the lead over the tree-based learners therefore belongs to a smooth score over well-scaled inputs: the thresholds of tree ensembles carve out the neighbourhood of each training positive, which is rewarded when held-out occurrences lie close by and penalised when they do not. The design of the network matters within this smooth model class (Appendix F, Table F11). Its score is a sum of a few ReLU ridges in the space of its inputs, so its resolution along each feature is set by how that feature is scaled; on z-scored, strongly skewed layers a handful of extreme values occupy most of the input range and the bulk of the ground is compressed into a narrow interval. Quantile-normal inputs give every feature a comparable, rank-based scale, the information that trees exploit through their thresholds, while keeping the score smooth, and background-heavy batches let the loss see much more of the background per update.

Third, the tests of the semi-supervised premise show when treating the background as unlabelled rather than barren pays off: when the evidence does not mark the known complexes, and when the background contains ground that resembles the occurrences. Without point-level map classes, DevNet beats its cross-entropy twin at every label count from one training positive to all of them, and bagging-PU beats RF. With the raw map classes, where six of the 15 Superior occurrences sit on an alkalic-intrusion cell that covers 0.5% of the background, a cross-entropy loss can exploit this near-deterministic lookup at least as well, and the two losses are level. A plausible reading is that the inlier term of the deviation loss, which pulls the scores of the whole background towards a common reference rather than pushing every background point down, regularises the score where the signal is weak and spread over many layers, whereas a supervised loss gains where one feature separates most positives from the background. Mislabelled ground around occurrences shows the premise most directly: labelling it barren cost RF almost half of its AP in Superior, whereas bagging-PU, which never treats any one background point as certainly barren, lost little, and across the three provinces the unlabelled-background member of each pair lost less in five of six comparisons and clearly more in none. A larger unlabelled pool did not favour the semi-supervised loss, contrary to our prediction.

The national results add a bound of scale. DevNet's deviation loss assumes that the unlabelled background is one normal population around which occurrences deviate. Within a province this works, but the national background mixes provinces that a random forest separates perfectly, and DevNet is then the weakest evidence-based model, below its own cross-entropy twin. The tests of Section 4.1.6 locate the failure: giving each province its own background reference doubles DevNet's AP, more than the same change helps RF, whereas a province indicator, which leaves the single reference in place, does not help at all. The pooling test points the same way: background from other provinces cost DevNet more than any other model. The semi-supervised premise, that unlabelled data describe the ground the occurrences deviate from, holds only for unlabelled data from the same geological setting; for a deviation loss that setting is a province, not a country.

These results matter for the use argued in Section 5.3: finding complexes that the bedrock map does not show is the setting without map classes, and there the semi-supervised formulation helps consistently. Among formulations that treat the background as unlabelled, the deviation loss is the one that works. nnPU, which differs from DevNet only in its risk, trails it by 0.19–0.41 in AP under blocking, fails in Cordillera and Grenville and without map classes, and depends on a class prior that is unknown in practice (AP 0.13–0.26 for π between 0.005 and 0.05); bagging-PU trails it under every protocol.

The Cordilleran and Grenville results bound the conclusion further. Where positives are comparatively plentiful and clustered (Cordillera), or the geophysical signal is weak (Grenville), the tree-based learners are as good or better under random splits, a coordinate-only forest matches or nearly matches them, and no model transfers between blocks. DevNet is therefore not a universal replacement for discriminative baselines; its advantage is specific to the combination of very few positives, a shared control and the need to extrapolate.

5.3 Geological map classes: evidence and lookup

The map classes move every model more than the choice of model does. Withholding them costs DevNet and RF about half or more of their AP, and a single class, alkalic intrusive rock, carries over a third of DevNet's blocked AP. Six of the 15 Superior occurrences lie on cells mapped as alkalic intrusive rock, which covers 0.5% of the background; for these occurrences the map already marks the target. The age-contact flag of the containing cell partly does the same, because a Proterozoic complex mapped within Archean crust creates an age contact, and DevNet ranks it second.

Whether such lookup is acceptable depends on the use. For targeting within mapped complexes, or for ranking known complexes by their geophysical setting, a mapped alkaline intrusion is legitimate and strong evidence, and the full inputs are appropriate. For finding complexes that the 1:5,000,000 map does not show, it is not, and the relevant skill is that without map classes: an AP of 0.08–0.13 under spatial blocking, 16–27 times the random-ranking value but well below what the full inputs suggest. Comparisons between studies, and claims about how well a model "discovers" occurrences, should state which of the two is being measured. In both settings DevNet is at or near the top under blocking, and without map classes its semi-supervised loss itself adds skill (Section 4.2.5); its absolute skill, however, depends strongly on the map classes.

5.4 Why the provinces differ

Grenville is the weakest of the three provinces for every model and Superior the strongest. The evidence layers are the same national compilations everywhere and almost complete (about 0.01% of values missing in each province), so the contrast does not reflect data coverage. It follows instead from the label sets and from the geology they sample (Table 20).

First, the effective number of independent occurrences is smaller than the count suggests, and smallest in Grenville. Merging occurrences within 10 km leaves 12 sites of 15 occurrences in Superior, 17 of 27 in Cordillera and 8 of 12 in Grenville, and in Cordillera and Grenville more than half of the occurrences share a site with another (median distance to the nearest other occurrence 7 km, against 31 km in Superior). Under a random split such neighbours fall on both sides, which is why coordinates alone score well in these two provinces (Section 4.2.6) and why their skill collapses under blocking.

Second, only in Superior do distant occurrences share a geophysical signature. There the far-neighbour index has a median of 0.005, an order of magnitude below that of random pseudo-occurrences (0.049; 5th percentile 0.022): an occurrence resembles another occurrence more than 100 km away more closely than it resembles all but a few background points. In Cordillera (0.032 against 0.027) and Grenville (0.046 against 0.066) the index falls within the range of the random sets, so there is no shared signature for a model to transfer. The univariate evidence agrees: the median feature separates occurrences from background with $|\mathrm{AUC} - 0.5| = 0.15$ in Superior but only 0.05 and 0.06 in Cordillera and Grenville, where three features and one feature respectively exceed 0.25.

Table 20 Province diagnostics. Sites: occurrences merged by single linkage at 10 km; shared: share of occurrences that share a site; NN: median distance to the nearest other occurrence; eras: number of geological-era classes mapped at the occurrences. $|\mathrm{AUC} - 0.5|$: median over features (occurrences against the evaluation background) and number of features with $|\mathrm{AUC} - 0.5| \geq 0.25$. Far-neighbour index: median over occurrences, geophysical and structural layers; random: median [5th percentile] of the same statistic for 200 random sets of background points.

	Label set					$\lvert\mathrm{AUC} - 0.5\rvert$		Far-neighbour index	
Region	Occ.	Sites	Shared	NN (km)	Eras	Median	≥ 0.25	Occurrences	Random
Superior	15	12	33%	31	2	0.15	6	0.005	0.049 [0.022]
Cordillera	27	17	56%	7	2	0.05	3	0.032	0.027 [0.015]
Grenville	12	8	58%	7	3	0.06	1	0.046	0.066 [0.031]

Third, the Grenville label set mixes carbonatites of different ages and tectonic settings. Crevier is a late-Grenvillian alkaline–carbonatite intrusion of about 957 Ma (Groulier et al., 2020). St-Honoré, with Terres Rompues on the same complex, was emplaced at about 580 Ma in the Saguenay graben of the St. Lawrence rift system (Néron, Bédard, & Gaboury, 2018). The Lavergne–Springer occurrences lie near Lake Nipissing at the western end of the Ottawa–Bonnechère graben, and the Cantley, Quinnville and Templeton showings near Gatineau belong to the calcite-rich veins and dykes of the Grenville marble belts, whose magmatic origin has long been debated (Moecher, Anderson, Cook, & Mezger, 1997). The far-neighbour index tracks this heterogeneity. The rift-related St-Honoré, Terres Rompues and Lavergne–Springer occurrences, about 700 km apart, resemble one another closely (indices 0.005–0.033), whereas Crevier (0.19), Megiscane Lake (0.28) and Sullivan Island (0.15) resemble no other occurrence. Age diversity alone is not the problem, since the Superior occurrences also span several magmatic episodes (Rukhlov & Bell, 2010). They were, however, all emplaced into the same stable cratonic lithosphere, whose margins and crustal-scale structures the present-day LAB, Moho and potential-field layers still record. The Grenville crust was reworked by the Grenvillian orogeny and by later rifting, so its present-day geophysics need not record a common setting for occurrences that formed in different ones. The mapped geology adds noise of its own: the twelve Grenville occurrences fall on Archean, Proterozoic and Paleozoic map units, because the bedrock map records reworked protoliths and the Paleozoic cover over St-Honoré rather than the carbonatites, and none of them lies on a cell mapped as alkalic intrusive rock, whereas about 5% of the Grenville background does.

Missing or coarse geophysics therefore explains little of the contrast; the number, independence and geological coherence of the labels explain most of it. A model for Grenville would need either more occurrences of one carbonatite type or separate models for rift-related and orogenic types, which the present label set cannot support.

5.5 Synthetic positives do not create information

Synthetic positives are an attractive response to label scarcity, and GAN-generated labels have been recommended for exactly this commodity (Parsa et al., 2025). Our

results counsel caution. A generator fitted to a dozen training positives has two options. It can stay close to them, in which case it adds near-duplicates that carry no new information (jitter and SMOTE, nearest-neighbour ratios of 0.1–0.2); or it can spread out, in which case it must guess where unseen occurrences lie and, having no information about that, places part of its mass in background-like feature space (WGAN-GP, 35% of samples nearest to a background point). Neither option supplies the information that is actually missing, namely what undiscovered occurrences look like. The gains for RF from jitter are better explained as smoothing: synthetic points around the training positives make more trees vote for nearby held-out points, which relieves RF's degenerate surface of exactly-zero probabilities rather than teaching it new geology. Consistent with this reading, they vanish and turn into losses with 200 km blocks, where held-out occurrences are far from every training one, and DevNet, whose loss already produces smooth scores, gains nothing from the same points. Where synthetic positives are used, they must be generated inside each training fold from that fold's positives only, and performance must be measured on real held-out occurrences; otherwise a synthetic copy of a training occurrence in the test set reproduces the sibling problem of Section C in another form.

5.6 The validation protocol decides many reported differences

Four of our findings concern evaluation rather than models. First, with tens of positives a single cross-validation split gives a noisy estimate: re-splitting the same Superior data moves each model's AP over a range of 0.07–0.17 (Table 9), as large as many of the differences between models. Second, random splits overstate skill wherever occurrences cluster, and the size of the overstatement differs between models, so random splits can also reverse a model ranking (Section 4.2.2). Third, rebuilt nationally, the S2-grid design gives a test AUC of 1.000, and two of its elements each suffice for it: splitting augmented child cells at random, because almost every test positive has a sibling, sharing the coarse deep-architecture values and the map classes exactly and the fine-scale values nearly, on the training side; and drawing negatives from the least similar ground, which gives 0.997–1.000 even without the sibling leak. In this design the held-out test asks the model to recognise a location it has already seen, and the comparison of training and test AUC, which Parsa et al. (2024) used to argue that their models were not over-fitted, becomes uninformative because the test set is not independent of the training set. Selecting negatives from the least similar 5% of the ground adds a second, independent source of optimism: it removes from evaluation exactly the ground on which a prospectivity model is hardest to get right. Fourth, the geological map classes can supply a large share of the reported skill by marking known complexes (Section 5.3), which no split design removes.

We stress what the ladder does and does not show. It uses our inputs, labels and models, not those of Parsa et al. (2024), and it does not establish what their numbers would have been under a grouped split. It does show that, on the national Canadian carbonatite data, the published design produces a test AUC of 1.000 while the same forest, once the leak and the easy negatives are removed, reaches an AP of 0.18 at a realistic prevalence and 0.03 under 100 km blocks, and an AP of 0.007 on Superior when Superior is held out. Neither the high AUC nor its small spread over splits can

be taken as evidence of generalisation to unexplored ground. The provincial version of the audit (Appendix C) shows the same in each province, where the published design gives test AUCs close to 1 whether the underlying skill is high (Superior) or modest (Grenville, honest AUC 0.77). Parsa et al. (2025) describe the same stratified random 70/30 split for a workflow that adds GAN-generated labels; their description does not state whether synthetic labels derived from an original label were kept on the same side of the split as that label. If they were not, the caveat applies with greater force. For prospectivity studies under extreme label scarcity we therefore recommend (i) grouping every augmented or near-duplicate label with its source occurrence before any split, (ii) drawing evaluation negatives at random from the domain rather than from the ground least similar to the positives, (iii) reporting AP with intervals over repeated splits rather than a single AUC, (iv) adding a spatially blocked validation in which training background is purged from held-out blocks, (v) reporting a geography-only baseline, so that readers can see how much of a score is location, (vi) reporting results with and without point-level map classes, so that readers can see how much of a score is map lookup, and (vii) for models pooled across geological provinces, testing transfer to held-out provinces and computing extent-dependent inputs within fixed geological units.

5.7 Geological interpretation

The drivers on which RF and DevNet agree in Superior describe a coherent setting. Shallower LAB, together with the strong ranking of deep-source magnetic gradients, points to the thinner lithosphere at the edges of the thick Superior root, where small-degree melting of metasomatised mantle and access along lithosphere-scale structures are most likely (Griffin et al., 2013; Groves & Santosh, 2021; Hoggard et al., 2020); a mantle conductor beneath the southern Abitibi is consistent with such metasomatism (Snyder et al., 2021). Fault proximity and high magnetic intensity capture the structures that focus ascent and the magnetite-bearing intrusions themselves (Thomas et al., 2016). The negative effect of proximity to deep magnetic worms is less intuitive; it may reflect the concentration of deep magnetic contacts along terrane boundaries that are not themselves carbonatite hosts, and should not be over-interpreted.

The strong positive effect of greenstone lithology and of proximity to regional greenstone units needs more care. Eight of the 15 occurrences belong to the eastern-Abitibi cluster in a greenstone-dominated subprovince, so the association may partly record where the largest cluster lies rather than a control shared by the dispersed occurrences; the dispersed occurrences that DevNet ranks lowest under blocking, Kapuskasing and Nemegosenda Lake, lie in the high-grade and alkaline terrain of the KSZ rather than in a greenstone belt. Consistent with this, DevNet's map follows the KSZ only in discontinuous patches, while the untrained Clay-Howells resource on the KSZ still ranks at its 97th percentile. A model whose top classes are tied to one greenstone-hosted cluster is useful for ranking ground of that type, but it should not be expected to recognise alkaline complexes in other settings as well.

5.8 Limitations

Fifteen positives allow only weak statistical statements: our firm results are those that are consistent across repeats, block sizes, comparisons and provinces. DevNet's input representation and batch composition were chosen with diagnostic experiments on these three provinces, and the nested selection shows that part of the final configuration's advantage reflects that choice; confirmation on an independent province is needed, and the Churchill Province, the obvious candidate, holds too few carbonatite-hosted occurrences after deduplication for a meaningful test. The geological encoding used here replaced a coarser grouping of lithologies after that grouping had been evaluated; the change raised the AP of every model except nnPU under random splits and 50–100 km blocks but not with 200 km blocks. The evaluation background may contain undiscovered occurrences, so all metrics are lower bounds. The test with background added near occurrences is a stress test: 20–50 points per occurrence far exceed the density of the training background, which places about one point in each 2–10 km annulus, so it shows sensitivity to mislabelled ground rather than its size in practice. Radiometric data could not be used because of incomplete coverage, and whole-rock geochemistry is too sparse to serve as an input. Eleven gravity and magnetic layers are proprietary Getech products; equivalent derivatives can be computed from the public parent grids, but results will differ somewhat. The provenance of the fault compilation should be confirmed against the original data release before publication. The occurrence-class information (Deposit/Prospect, Showing, Occurrence) was not used for weighting; with 3–8 occurrences per class, confidence-weighted training could not be evaluated reliably. Finally, maps are per-region percentiles and are not calibrated probabilities. At the national scale, 12 of the 66 occurrences lie outside the three study provinces and four of them form one cluster (Ashram), so the "rest of Canada" group is small and spatially dependent; the national labels come from two compilations: a third (Huston, 2024), used by a parallel national study, would add one deposit (Hoidas Lake), and the five records of Orris et al. (2018) whose carbonatite affinity is marked as uncertain, which we excluded, would add five more (one of them Clay-Howells, our external check); the 0.1° lattice samples high latitudes more densely per unit area; and the national DevNet uses the capacity chosen for Superior and Grenville.

6 Conclusions

For carbonatite-hosted REE prospectivity mapping in Canada, with 66 occurrences nationally and 12–27 per province:

1. Under the same leakage-free validation as within the provinces, national models reach an AP of at most 0.06 with 100 km blocks (random ranking 0.003). A province's model ranks another province's occurrences at or below chance in most pairs; the provinces are separable domains, and several inputs mark occurrences in opposite directions in different provinces. Pooling the country into a province's training data lowered its blocked AP in 11 of 15 model–province combinations, and in Superior a model trained on Superior alone remains best.

2. Rebuilt on these data, the national S2-grid design of Parsa et al. (2024) gives a test AUC of 1.000, and random splitting of augmented child cells and negatives drawn from the least similar ground each suffice for it. With both removed, the same forest reaches an AP of 0.18 at realistic prevalence and 0.03 under 100 km blocks.
3. Inputs whose values depend on the extent of the study area, such as quantised textures and generalised maps, must be computed within fixed geological units before scales are compared; computed over the whole country they lowered Superior DevNet's blocked AP from 0.343 to 0.291 before any model changed.
4. In the Superior Province a semi-supervised Deviation Network ranks held-out occurrences better than Random Forest, XGBoost, tuned Random Forest and gradient boosting, bagging-PU and nnPU under random and 50–200 km spatially blocked cross-validation (AP 0.45 and 0.42–0.25 against 0.27–0.28 and 0.20–0.12 for RF and XGBoost), in every repeat except one against XGBoost; with 15 occurrences the margin over RF is consistent but not statistically conclusive.
5. DevNet's lead over the tree-based learners comes mainly from a smooth score over rank-normal inputs with background-heavy batches: with the full inputs, an ablation that labels the background barren (cross-entropy) does as well. Treating the background as unlabelled pays off when the evidence does not mark the known complexes: without point-level map classes DevNet beats its cross-entropy twin at every label count (15–20 of 20 repeats) and bagging-PU beats RF, and labelling ground around occurrences barren almost halves RF's AP but barely affects bagging-PU, a pattern that recurs in all three provinces. Nationally DevNet is the weakest evidence-based model; a background reference per province doubles its AP, so the deviation loss's premise of one normal background holds within a province, not across a country.
6. Geological map classes move every model more than the choice of model. A single mapped alkalic-intrusion class, covering 6 of 15 Superior occurrences and 0.5% of the background, carries over a third of DevNet's blocked AP, and without point-level map classes DevNet and RF lose about half or more of their skill. Prospectivity studies should report results with and without such classes.
7. In the Cordilleran Orogen and the Grenville Province the tree-based learners lead under random splits, a forest given only coordinates matches them in Cordillera and comes close in Grenville, and no model transfers between blocks; the number, independence and geological coherence of the labels, not data coverage, explain the contrast.
8. Synthetic positives from jitter, SMOTE or a tabular WGAN-GP, generated strictly within training folds, gave DevNet no gain and RF only gains that vanish at 200 km.
9. RF and DevNet agree on the direction of all 30 leading drivers across the three provinces. DevNet's Superior map is more stable than RF's when occurrences are left out, places the untrained Clay-Howells Fe–REE–Nb resource at its 97th percentile, and favours greenstone-hosted settings like the dominant Abitibi cluster.

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Declarations

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Competing interests. The authors declare no competing interests.

Data availability. Occurrence labels are derived from Orris et al. (2018) and Parsa et al. (2024); the deduplicated provincial and national tables and the study-area outlines are distributed with the code. Public layers are available from McCafferty et al. (2023), Jobin et al. (2017), Miles and Oneschuk (2016), D.C. Bradley (2008) and Wheeler et al. (1996). Eleven processed gravity and magnetic grids were supplied by Getech Group plc under licence and cannot be redistributed.

Code availability. The pipeline, every analysis in this paper and the scripts that generate its tables and figures are available at https://github.com/miaozhou-312/ree-prospectivity-paper.

Appendix A Additional results

Table A1 AUC (median over six repeats) for the runs of Table 11. Bold: highest AUC in each row.

Region	Split	DevNet	RF	XGBoost	Bagging-PU	nnPU
Superior	random	0.972	**0.983**	0.976	0.976	0.838
	50 km	**0.973**	0.963	0.966	0.965	0.824
	100 km	**0.950**	0.904	0.922	0.945	0.832
	200 km	0.783	**0.880**	0.826	0.855	0.814
Cordillera	random	0.914	0.931	0.927	**0.936**	0.553
	50 km	0.863	0.849	0.852	**0.866**	0.545
	100 km	0.790	0.776	0.803	**0.847**	0.541
	200 km	**0.821**	0.740	0.783	0.810	0.550
Grenville	random	0.799	0.838	0.857	**0.898**	0.542
	50 km	0.657	0.736	0.709	**0.780**	0.536
	100 km	0.661	0.538	0.621	**0.687**	0.568
	200 km	0.568	0.557	0.584	**0.671**	0.565

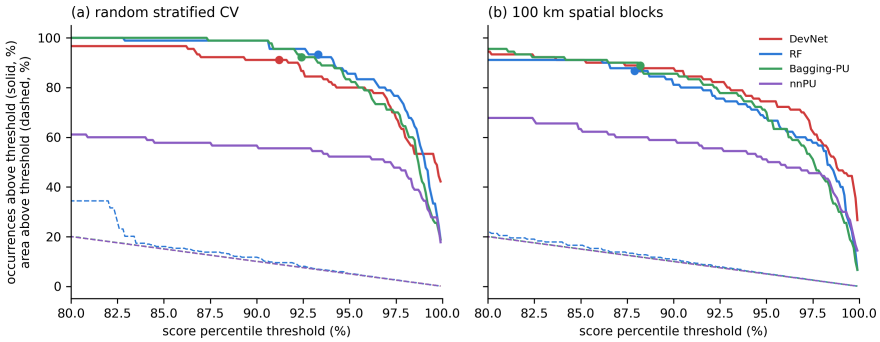


Fig. A1 Prediction–area curves for Superior from out-of-fold ranks: share of held-out occurrences (solid) and of the evaluation background, a proxy for area (dashed), above a score-percentile threshold; dots mark the intersections. (a) Random cross-validation, (b) 100 km spatial blocks.

Table A2 Nested design selection: AP (median over six repeats) when DevNet's input transformation and batch composition are chosen inside each outer training fold by inner 100 km spatial cross-validation, compared with the final configuration and with RF. Δ: paired difference on repeat-averaged ranks (repeats won). Chosen: outer folds choosing quantile 16+240 / quantile 16+16 / z-score 16+240 / z-score 16+16.

Region	Split	Nested	Final	Δ vs final	RF	Δ vs RF	Chosen
Superior	random	0.354	0.445	-0.055	0.271	+0.155 (5/6)	10/4/11/5
	50 km	0.335	0.418	-0.046	0.192	+0.194 (6/6)	12/2/10/6
	100 km	0.312	0.343	+0.023	0.168	+0.219 (6/6)	9/2/13/6
	200 km	0.190	0.252	-0.064	0.151	+0.120 (6/6)	11/1/10/8
Cordillera	random	0.335	0.341	+0.010	0.427	-0.083 (0/6)	12/7/5/6
	50 km	0.133	0.129	+0.057	0.092	+0.068 (5/6)	11/5/7/7
	100 km	0.069	0.055	-0.004	0.046	-0.034 (4/6)	7/5/8/10
	200 km	0.062	0.076	+0.000	0.039	+0.000 (5/6)	5/7/7/11
Grenville	random	0.116	0.125	+0.041	0.257	-0.217 (0/6)	16/10/1/3
	50 km	0.021	0.031	-0.002	0.086	-0.003 (0/6)	20/6/1/3
	100 km	0.007	0.007	-0.002	0.008	-0.004 (2/6)	16/6/2/6
	200 km	0.005	0.005	-0.001	0.006	-0.001 (0/6)	17/5/1/6

Table A3 Sensitivity of nnPU to the assumed class prior π, Superior (random cross-validation; median and 95% interval over six repeats).

π	AUC	AP
0.005	0.811 [0.779, 0.861]	0.133 [0.099, 0.208]
0.01	0.838 [0.794, 0.892]	0.152 [0.125, 0.191]
0.02	0.869 [0.825, 0.916]	0.182 [0.143, 0.263]
0.05	0.902 [0.885, 0.924]	0.264 [0.179, 0.311]

Appendix B National scale: additional results

Table B4 Why national inputs first scored the provinces lower: AP (median over six repeats) of each province's models trained and tested on its own data with four input tables. A: the province's inputs and values (the main analysis). B: the province's inputs with GLCM textures and generalised-geology distances computed over the whole of Canada. C: inputs pruned on the national table, values computed over Canada. D: inputs pruned on the national table, provincial values. All other inputs take identical values in all four. The national analyses use per-province values, for which B equals A and D equals C.

Province	Model	Split	A	B	C	D
Superior	DevNet	random	0.445	0.323	0.311	0.356
	DevNet	100 km	0.343	0.291	0.238	0.307
	RF	random	0.271	0.293	0.289	0.295
	RF	100 km	0.168	0.172	0.143	0.133
Cordillera	DevNet	random	0.341	0.276	0.246	0.315
	DevNet	100 km	0.055	0.071	0.066	0.049
	RF	random	0.427	0.457	0.463	0.436
	RF	100 km	0.046	0.079	0.080	0.055
Grenville	DevNet	random	0.125	0.070	0.083	0.110
	DevNet	100 km	0.007	0.005	0.005	0.007
	RF	random	0.257	0.178	0.218	0.243
	RF	100 km	0.008	0.008	0.008	0.008

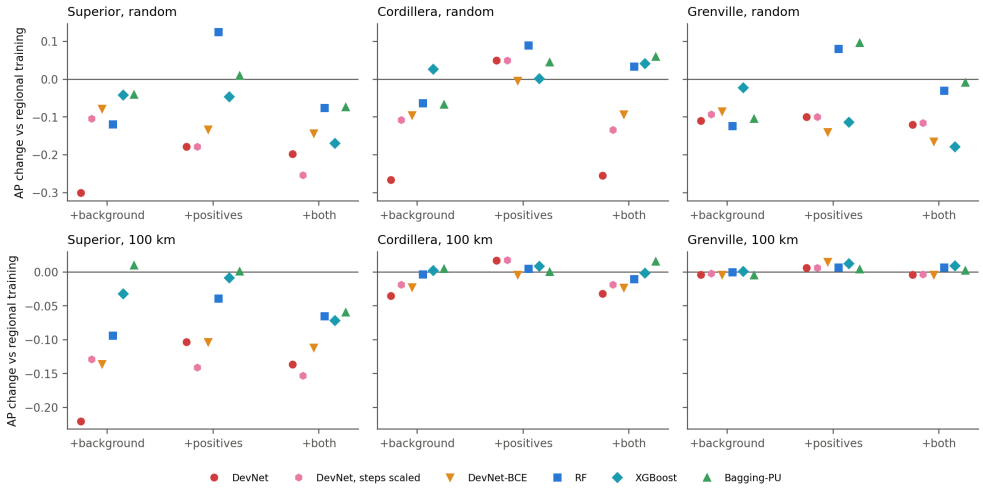


Fig. B2 Pooling the country into a province's training data: change in AP from the province's own data when the national training background outside it, the national occurrences outside it, or both are added (median over six repeats; top: random cross-validation, bottom: 100 km blocks).

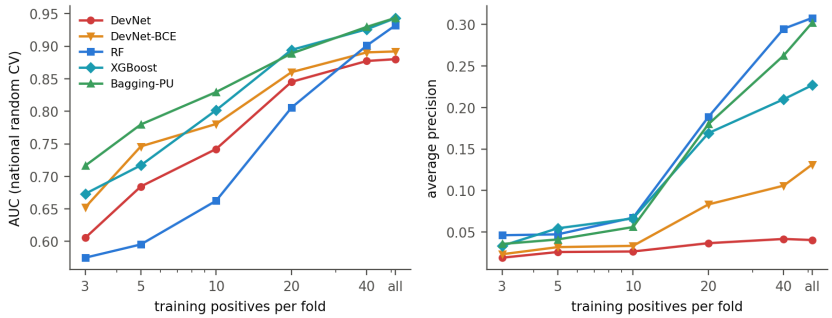


Fig. B3 National learning curve: AUC and AP under random cross-validation when only k training occurrences per fold are kept (median over six repeats).

Table B5 National learning curve: AP (median over six repeats) under random cross-validation with k training occurrences per fold.

k	DevNet	DevNet-BCE	RF	XGBoost	Bagging-PU
3	0.019	0.023	0.046	0.033	0.036
5	0.026	0.032	0.047	0.055	0.041
10	0.027	0.033	0.067	0.066	0.056
20	0.037	0.083	0.189	0.169	0.180
40	0.042	0.106	0.295	0.210	0.263
52 (all)	0.040	0.131	0.308	0.227	0.302

Appendix C Provincial protocol audit

To quantify how the validation design of Parsa et al. (2024) behaves on these data, we rebuilt it with the same S2 geometry library. Each occurrence's level-12 S2 cell (about 4.5 km^2) was replaced by its four level-13 child cells, and all inputs were extracted at the child-cell centres with the same pipeline. An equal number of negatives was drawn either from the 5% of background candidates with the lowest mean cosine similarity to the augmented positives in min–max scaled feature space (as described by Parsa et al., 2024) or at random. Random forests of 1,000 trees on min–max scaled features were evaluated with 50 stratified 70/30 splits. Three split designs were compared: (i) child-level random splitting, as described in the original study; (ii) splitting with all children of one parent kept on the same side; and (iii) no augmentation, one label per occurrence. We report mean test AUC, its 2.5–97.5 percentile range over the 50 splits, and the share of test positives with a sibling in the training set. The audit uses our inputs, regions and models rather than those of the original study, so it measures the sensitivity of the protocol, not the original study's numbers.

Rebuilding the S2 parent–child augmentation and child-level random splitting of Parsa et al. (2024) on our data gives a test AUC of 0.999–1.000 in all three provinces and for both negative-sampling rules (Table C7; Fig. C4). In 97–99% of the test positives, at least one sibling child cell of the same occurrence is in the training set. In feature space, siblings are closer to each other (median standardised distance 2.5–3.9) than occurrences are to their nearest other occurrence (4.3–7.1) and than background points are to their nearest background neighbour (4.6–4.9; Table C6), because the coarse deep-architecture layers and the map classes take identical values in sibling cells and the fine-scale layers nearly so.

The honest reference within this balanced 70/30 design is one label per occurrence with random negatives. It gives a mean AUC of 0.976 in Superior, 0.887 in Cordillera and 0.767 in Grenville, each with a wide range over the 50 splits (0.80–1.00, 0.75–1.00 and 0.50–1.00). The child-level split therefore inflates test AUC by about 0.02, 0.11 and 0.23 respectively, the inflation being largest where real skill is lowest, and it collapses the apparent uncertainty to almost nothing. Drawing negatives from the least similar 5% of the ground inflates AUC independently of the split: without augmentation, cosine-dissimilar negatives raise mean AUC to 0.999, 0.997 and 1.000. Keeping all children of one parent on the same side of the split removes most, but not all, of the augmentation effect (Superior 0.972, Cordillera 0.961, Grenville 0.905 with random negatives); the remainder reflects the fourfold larger training set and spatial dependence between neighbouring occurrences in different parent cells, which only spatial blocking removes.

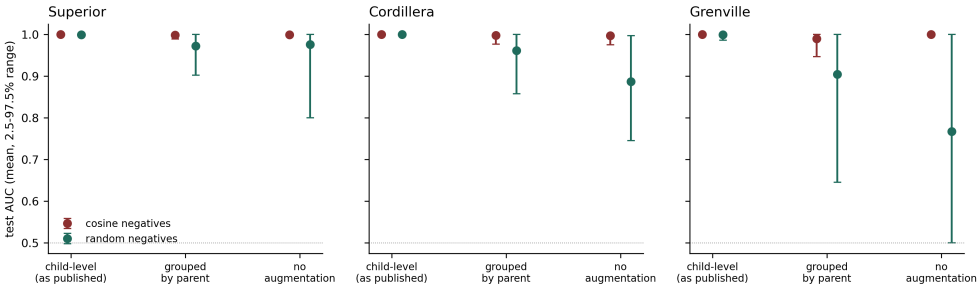


Fig. C4 Protocol audit: random forest test AUC under S2 parent–child augmentation for three split designs and two negative-sampling rules (mean and 2.5–97.5 percentile range over 50 balanced 70/30 splits).

Table C6 Feature-space distance (median, standardised features) between sibling child cells of the same occurrence, from each occurrence to its nearest other occurrence, and from background points to their nearest background neighbour.

Region	Siblings	Nearest other occurrence	Nearest background neighbour
Superior	3.89	5.36	4.59
Cordillera	2.53	4.29	4.68
Grenville	3.25	7.09	4.90

Table C7 Protocol audit: random forest test AUC (mean and 2.5–97.5 percentile range over 50 stratified 70/30 splits of balanced labels) under the S2 parent–child augmentation of Parsa et al. (2024), for two negative-sampling rules (cosine-dissimilar or random) and three split designs (child-level random, as published; children grouped by parent; no augmentation). Pos.: positive labels. Sibling: share of test positives with a sibling child cell in the training set.

Region	Negatives	Split	Pos.	Test AUC	Sibling
Superior	cosine	child-level	60	1.000 [1.000, 1.000]	98%
	cosine	parent-grouped	60	0.999 [0.989, 1.000]	–
	cosine	no augm.	15	0.999 [1.000, 1.000]	–
	random	child-level	60	0.999 [0.995, 1.000]	98%
	random	parent-grouped	60	0.972 [0.902, 1.000]	–
	random	no augm.	15	0.976 [0.800, 1.000]	–
Cordillera	cosine	child-level	108	1.000 [1.000, 1.000]	97%
	cosine	parent-grouped	108	0.998 [0.976, 1.000]	–
	cosine	no augm.	27	0.997 [0.975, 1.000]	–
	random	child-level	108	1.000 [0.998, 1.000]	97%
	random	parent-grouped	108	0.961 [0.858, 1.000]	–
	random	no augm.	27	0.887 [0.745, 0.997]	–
Grenville	cosine	child-level	48	1.000 [1.000, 1.000]	99%
	cosine	parent-grouped	48	0.990 [0.946, 1.000]	–
	cosine	no augm.	12	1.000 [1.000, 1.000]	–
	random	child-level	48	0.999 [0.986, 1.000]	99%
	random	parent-grouped	48	0.905 [0.646, 1.000]	–
	random	no augm.	12	0.767 [0.500, 1.000]	–

Appendix D Augmentation under spatially blocked validation

Table D8 Superior average precision with fold-safe synthetic positives under spatially blocked cross-validation (median and 95% interval over six repeats), with the paired per-repeat difference from no augmentation.

Blocks	Model	Augmentation	AP	ΔAP per repeat	wins
100 km	DevNet	none	0.343 [0.184, 0.452]	–	–
		Gaussian jitter 10×	0.336 [0.163, 0.456]	-0.000 [-0.024, +0.005]	3/6
		SMOTE 3×	0.346 [0.166, 0.448]	-0.006 [-0.020, +0.019]	2/6
		WGAN-GP 3×	0.342 [0.220, 0.541]	-0.005 [-0.063, +0.201]	3/6
		WGAN-GP 10×	0.321 [0.238, 0.395]	-0.031 [-0.095, +0.066]	2/6
100 km	RF	none	0.168 [0.067, 0.181]	–	–
		Gaussian jitter 10×	0.157 [0.070, 0.246]	+0.004 [-0.032, +0.072]	5/6
		SMOTE 3×	0.154 [0.059, 0.177]	-0.006 [-0.027, +0.005]	1/6
		WGAN-GP 3×	0.175 [0.093, 0.224]	+0.033 [-0.031, +0.051]	5/6
		WGAN-GP 10×	0.219 [0.091, 0.273]	+0.050 [+0.004, +0.101]	6/6
200 km	DevNet	none	0.252 [0.169, 0.342]	–	–
		Gaussian jitter 10×	0.247 [0.167, 0.336]	-0.005 [-0.009, +0.004]	2/6
		SMOTE 3×	0.246 [0.162, 0.365]	-0.003 [-0.014, +0.027]	2/6
		WGAN-GP 3×	0.260 [0.172, 0.334]	+0.001 [-0.027, +0.085]	3/6
		WGAN-GP 10×	0.248 [0.244, 0.317]	-0.006 [-0.035, +0.080]	2/6
200 km	RF	none	0.151 [0.106, 0.158]	–	–
		Gaussian jitter 10×	0.100 [0.076, 0.129]	-0.047 [-0.077, +0.014]	2/6
		SMOTE 3×	0.104 [0.055, 0.122]	-0.041 [-0.064, -0.005]	0/6
		WGAN-GP 3×	0.122 [0.089, 0.183]	-0.019 [-0.049, +0.042]	2/6
		WGAN-GP 10×	0.137 [0.093, 0.171]	-0.001 [-0.060, +0.049]	3/6

Appendix E Interpretation in the contrasting provinces

Table E9 As Table 19, Cordillera (55 features).

Feature	DevNet rank	DevNet dir.	RF rank	RF dir.
Distance to regional high-grade metam.	4	↓ (-0.97)	1	↓ (-0.86)
Era: Mesozoic	3	↓ (-0.82)	14	↓ (-0.82)
Bouguer anomaly (BA)	13	↓ (-0.96)	3	↓ (-0.87)
Era: Proterozoic	2	↑ (+0.67)	18	↑ (+0.70)
Age contact (H3 cell)	1	↑ (+0.85)	34	↑ (+0.85)
Moho depth	5	↑ (+0.96)	10	↑ (+0.89)
Passive-margin proximity	32	↑ (+0.73)	2	↑ (+0.78)
Distance to regional age contact	6	↑ (+0.96)	11	↑ (+0.90)
DRTP mean (10 km)	8	↓ (-0.95)	7	↓ (-0.84)
Distance to regional greenstone	30	↑ (+0.93)	4	↑ (+0.20)

Table E10 As Table 19, Grenville (58 features).

Feature	DevNet rank	DevNet dir.	RF rank	RF dir.
Deep-source RTP	1	↑ (+0.97)	2	↑ (+0.73)
Distance to regional age contact	4	↓ (-0.99)	1	↓ (-0.77)
DRTP THD tilt	9	↓ (-0.98)	4	↓ (-0.80)
Distance to regional mafic-ultramafic	15	↑ (+0.98)	3	↑ (+0.45)
Distance to regional sedimentary	11	↓ (-0.98)	5	↓ (-0.91)
BA high-pass 40 km std. dev. (10 km)	3	↑ (+0.98)	8	↑ (+0.85)
DRTP mean (10 km)	7	↑ (+0.97)	14	↑ (+0.94)
Deep gravity-worm proximity	2	↑ (+0.98)	23	↑ (+0.80)
Deep magnetic-worm proximity	6	↑ (+0.98)	25	↑ (+0.80)
Moho depth	20	↓ (-0.96)	6	↓ (-0.68)

Appendix F Ablation of DevNet design choices

DevNet's two design choices (Section 3.3) are examined here with the inputs of the main analysis, by changing one element at a time from a configuration with z-scored inputs and label-balanced batches of 16 + 16 rows. All configurations are evaluated with the folds, protocols and seeds of the main results (Table F11). In Superior, quantile inputs alone raise AP under random splits from 0.358 to 0.424, and background-heavy batches alone to 0.395; together they give 0.445, the highest AP under every protocol except 100 km blocks, where the z-scored, background-heavy variant is level (0.349 against 0.343). Training for longer, by contrast, repeats the dozen positives and pushes the network towards memorising their neighbourhoods: with z-scored inputs, ten times more steps left random-split AP unchanged (0.361 against 0.358) but lowered it under 100 km blocks (0.277 against 0.329). The nested early-stopping configuration chooses the number of steps (100–3,000, cosine schedule over 3,000 steps) inside each outer training fold by a 100 km spatial cross-validation (three inner folds, two inner restarts, background purged from inner validation blocks), so the test fold is never used for the choice. Table F12 gives the sensitivity of the final configuration to the deviation margin: smaller margins gave somewhat higher AP in all three provinces (Superior 0.469 with $a = 2$ against 0.445 with $a = 5$), but we keep the value of Pang et al. (2019) rather than tune it on these folds.

Table F11 Ablation of DevNet's design choices: AP (median over six repeats) under random cross-validation and spatial blocks, on identical folds. Inputs: quantile-normal or z-scored; batch: positive + background rows per step; steps: number of gradient steps (nested ES: chosen by nested 100 km spatial cross-validation). In brackets: number of repeats in which the final configuration (bold) has the higher AP. –: not run.

Region	Inputs	Batch	Steps	random	50 km	100 km	200 km
Superior	quantile	16 + 240	300 (final)	**0.445**	**0.418**	**0.343**	**0.252**
	z-score	16 + 240	300	0.395 (4/6)	0.380 (5/6)	0.349 (3/6)	0.221 (3/6)
	quantile	16 + 16	300	0.424 (5/6)	0.347 (6/6)	0.310 (5/6)	0.239 (5/6)
	z-score	16 + 16	300	0.358 (6/6)	0.332 (5/6)	0.329 (4/6)	0.226 (3/6)
	z-score	16 + 16	3,000	0.361 (6/6)	–	0.277 (4/6)	–
	z-score	16 + 16	nested ES	0.367 (6/6)	0.307 (5/6)	0.262 (4/6)	0.218 (6/6)
Cordillera	quantile	16 + 240	300 (final)	**0.341**	**0.129**	**0.055**	**0.076**
	z-score	16 + 240	300	0.297 (5/6)	0.132 (4/6)	0.078 (3/6)	0.065 (4/6)
	quantile	16 + 16	300	0.301 (6/6)	0.124 (6/6)	0.047 (5/6)	0.074 (2/6)
	z-score	16 + 16	300	0.271 (6/6)	0.154 (4/6)	0.085 (3/6)	0.067 (4/6)
	z-score	16 + 16	3,000	0.228 (6/6)	–	0.060 (3/6)	–
	z-score	16 + 16	nested ES	0.279 (6/6)	0.128 (5/6)	0.068 (4/6)	0.057 (4/6)
Grenville	quantile	16 + 240	300 (final)	**0.125**	**0.031**	**0.007**	**0.005**
	z-score	16 + 240	300	0.072 (6/6)	0.023 (5/6)	0.005 (6/6)	0.004 (6/6)
	quantile	16 + 16	300	0.122 (3/6)	0.030 (5/6)	0.006 (6/6)	0.005 (6/6)
	z-score	16 + 16	300	0.059 (6/6)	0.019 (6/6)	0.006 (6/6)	0.004 (6/6)
	z-score	16 + 16	3,000	0.105 (4/6)	–	0.005 (5/6)	–
	z-score	16 + 16	nested ES	0.088 (4/6)	0.028 (3/6)	0.004 (6/6)	0.004 (5/6)

Table F12 Sensitivity of DevNet to the deviation margin a (AP, median over six repeats, random cross-validation). The main results use $a = 5$.

Region	$a = 2$	$a = 3$	$a = 5$	$a = 8$
Superior	0.469	0.453	0.445	0.436
Cordillera	0.367	0.369	0.341	0.313
Grenville	0.183	0.156	0.125	0.120

References

Agterberg, F.P., & Bonham-Carter, G.F. (2005). Measuring the performance of mineral-potential maps. *Natural Resources Research*, *14*(1), 1–17, https://doi.org/10.1007/s11053-005-4674-0

Airola, A., Pohjankukka, J., Torppa, J., Middleton, M., Nykänen, V., Heikkonen, J., Pahikkala, T. (2019). The spatial leave-pair-out cross-validation method for reliable AUC estimation of spatial classifiers. *Data Mining and Knowledge Discovery*, *33*(3), 730–747, https://doi.org/10.1007/s10618-018-00607-x

Aranha, M., Porwal, A., Sundaralingam, M., González-Álvarez, I. (2021). Detection and extraction of geophysical features associated with carbonatites and alkaline complexes. *First international meeting for applied geoscience & energy expanded abstracts* (pp. 894–899).

Arjovsky, M., Chintala, S., Bottou, L. (2017). Wasserstein generative adversarial networks. *Proceedings of the 34th international conference on machine learning* (Vol. 70, pp. 214–223).

Bekker, J., & Davis, J. (2020). Learning from positive and unlabeled data: A survey. *Machine Learning*, *109*(4), 719–760, https://doi.org/10.1007/s10994-020-05877-5

Bradley, A.P. (1997). The use of the area under the ROC curve in the evaluation of machine learning algorithms. *Pattern Recognition*, *30*(7), 1145–1159, https://doi.org/10.1016/S0031-3203(96)00142-2

Bradley, D.C. (2008). Passive margins through earth history. *Earth-Science Reviews*, *91*(1–4), 1–26, https://doi.org/10.1016/j.earscirev.2008.08.001

Breiman, L. (2001). Random forests. *Machine Learning*, *45*(1), 5–32, https://doi.org/10.1023/A:1010933404324

Chawla, N.V., Bowyer, K.W., Hall, L.O., Kegelmeyer, W.P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, *16*, 321–357, https://doi.org/10.1613/jair.953

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery*

and data mining (pp. 785–794).

Chen, Y., & Wu, W. (2019). Isolation forest as an alternative data-driven mineral prospectivity mapping method with a higher data-processing efficiency. *Natural Resources Research*, *28*(1), 31–46, https://doi.org/10.1007/s11053-018-9375-6

Chung, C.-J.F., & Fabbri, A.G. (2003). Validation of spatial prediction models for landslide hazard mapping. *Natural Hazards*, *30*(3), 451–472, https://doi.org/10.1023/B:NHAZ.0000007172.62651.2b

Coolbaugh, M.F., Raines, G.L., Zehner, R.E. (2007). Assessment of exploration bias in data-driven predictive models and the estimation of undiscovered resources. *Natural Resources Research*, *16*(2), 199–207, https://doi.org/10.1007/s11053-007-9037-6

Cooper, G.R.J., & Cowan, D.R. (2005). The use of textural analysis to locate features in geophysical data. *Computers & Geosciences*, *31*(7), 882–890, https://doi.org/10.1016/j.cageo.2005.02.001

Dormann, C.F., Elith, J., Bacher, S., Buchmann, C., Carl, G., Carré, G., … Lautenbach, S. (2013). Collinearity: A review of methods to deal with it and a simulation study evaluating their performance. *Ecography*, *36*(1), 27–46, https://doi.org/10.1111/j.1600-0587.2012.07348.x

Drenth, B.J. (2014). Geophysical expression of a buried niobium and rare earth element deposit: The Elk Creek carbonatite, Nebraska, USA. *Interpretation*, *2*(4), SJ23–SJ33, https://doi.org/10.1190/INT-2014-0002.1

Efron, B., & Tibshirani, R.J. (1993). *An introduction to the bootstrap*. New York: Chapman & Hall.

Elkan, C., & Noto, K. (2008). Learning classifiers from only positive and unlabeled data. *Proceedings of the 14th acm sigkdd international conference on knowledge discovery and data mining* (pp. 213–220).

Gittins, J., Macintyre, R.M., York, D. (1967). The ages of carbonatite complexes in eastern Canada. *Canadian Journal of Earth Sciences*, *4*(4), 651–655, https://doi.org/10.1139/e67-042

Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., … Bengio, Y. (2014). Generative adversarial nets. *Advances in neural information processing systems 27* (pp. 2672–2680).

Gorishniy, Y., Rubachev, I., Khrulkov, V., Babenko, A. (2021). Revisiting deep learning models for tabular data. *Advances in neural information processing systems 34* (pp. 18932–18943).

Griffin, W.L., Begg, G.C., O'Reilly, S.Y. (2013). Continental-root control on the genesis of magmatic ore deposits. *Nature Geoscience*, *6*(11), 905–910, https://doi.org/10.1038/ngeo1954

Grinsztajn, L., Oyallon, E., Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data? *Advances in neural information processing systems 35, datasets and benchmarks track* (pp. 507–520).

Groulier, P.-A., et al. (2020). Silicate–carbonate liquid immiscibility: Insights from the Crevier alkaline intrusion (Quebec). *Journal of Petrology*, *61*(3), egaa033, https://doi.org/10.1093/petrology/egaa033

Groves, D.I., & Santosh, M. (2021). Craton and thick lithosphere margins: The sites of giant mineral deposits and mineral provinces. *Gondwana Research*, *100*, 195–222, https://doi.org/10.1016/j.gr.2020.06.008

Gulrajani, I., Ahmed, F., Arjovsky, M., Dumoulin, V., Courville, A. (2017). Improved training of Wasserstein GANs. *Advances in neural information processing systems 30* (pp. 5767–5777).

Hall-Beyer, M. (2017). Practical guidelines for choosing GLCM textures to use in landscape classification tasks over a range of moderate spatial scales. *International Journal of Remote Sensing*, *38*(5), 1312–1338, https://doi.org/10.1080/01431161.2016.1278314

Heather, K.B., Percival, J.A., Moser, D., Bleeker, W. (1995). *Tectonics and metallogeny of Archean crust in the Abitibi–Kapuskasing–Wawa region* (Open File No. 3141). Geological Survey of Canada.

Hoggard, M.J., Czarnota, K., Richards, F.D., Huston, D.L., Jaques, A.L., Ghelichkhan, S. (2020). Global distribution of sediment-hosted metals controlled by craton edge stability. *Nature Geoscience*, *13*(7), 504–510, https://doi.org/10.1038/s41561-020-0593-2

Hornby, P., Boschetti, F., Horowitz, F.G. (1999). Analysis of potential field data in the wavelet domain. *Geophysical Journal International*, *137*(1), 175–196, https://doi.org/10.1046/j.1365-246x.1999.00788.x

Hou, Z.-Q., Xu, B., Zhang, H., Zheng, Y.-C., Wang, R., Liu, Y., … O'Reilly, S.Y. (2023). Refertilized continental root controls the formation of the Mianning–Dechang carbonatite-associated rare-earth-element ore system. *Communications Earth & Environment*, *4*(1), 293, https://doi.org/10.1038/s43247-023-00956-6

Huston, D. (2024). *Global rare earth element deposits – a compilation.* Geoscience Australia, Canberra.

Jobin, D.M., Véronneau, M., Miles, W. (2017). *Gravity anomaly map, Canada* (Canadian Geoscience Map No. 8081). Geological Survey of Canada.

Kingma, D.P., & Ba, J. (2014). *Adam: A method for stochastic optimization.* arXiv:1412.6980.

Kiryo, R., Niu, G., du Plessis, M.C., Sugiyama, M. (2017). Positive-unlabeled learning with non-negative risk estimator. *Advances in neural information processing systems 30* (pp. 1675–1685).

Lundberg, S.M., Erion, G., Chen, H., DeGrave, A., Prutkin, J.M., Nair, B., … Lee, S.-I. (2020). From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, *2*(1), 56–67, https://doi.org/10.1038/s42256-019-0138-9

Lundberg, S.M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in neural information processing systems 30* (pp. 4765–4774).

Luo, Z., Farahbakhsh, E., Hore, S., Müller, R.D. (2026). DEEP-SEAM: An explainable semi-supervised deep learning framework for mineral prospectivity mapping. *Geoscientific Model Development*, *19*(7), 2593–2625, https://doi.org/10.5194/gmd-19-2593-2026

McCafferty, A.E., San Juan, C.A., Lawley, C., Graham, G.E., Gadd, M.G., Huston, D.L., … Czarnota, K. (2023). *National-scale geophysical, geologic, and mineral resource data and grids for the United States, Canada, and Australia: Data in support of the Tri-National Critical Minerals Mapping Initiative (ver. 1.1, march 2025).* U.S. Geological Survey data release.

Meyer, H., & Pebesma, E. (2021). Predicting into unknown space? estimating the area of applicability of spatial prediction models. *Methods in Ecology and Evolution*, *12*(9), 1620–1633, https://doi.org/10.1111/2041-210X.13650

Miles, W., & Oneschuk, D. (2016). *Magnetic anomaly map, Canada* (Canadian Geoscience Map No. 7799). Geological Survey of Canada.

Moecher, D.P., Anderson, E.D., Cook, C.A., Mezger, K. (1997). The petrogenesis of metamorphosed carbonatites in the Grenville Province, Ontario. *Canadian Journal of Earth Sciences*, *34*(9), 1185–1201, https://doi.org/10.1139/e17-095

Montsion, R.M., Thurston, P., Ayer, J. (2018). *1:2 000 000 scale geological compilation of the Superior Craton – version 1* (Tech. Rep. No. MERC-ME-2018-017). Sudbury, Ontario: Mineral Exploration Research Centre, Harquail School of Earth Sciences, Laurentian University.

Mordelet, F., & Vert, J.-P. (2014). A bagging SVM to learn from positive and unlabeled examples. *Pattern Recognition Letters*, *37*, 201–209, https://doi.org/10.1016/j.patrec.2013.06.010

Néron, A., Bédard, L.P., Gaboury, D. (2018). The Saint-Honoré carbonatite REE zone, Québec, Canada: Combined magmatic and hydrothermal processes. *Minerals*, *8*(9), 397, https://doi.org/10.3390/min8090397

Ontario Geological Survey (2011). *Mineral deposit inventory record MDI42G16SE00006: Clay-Howells.* Ontario Ministry of Northern Development and Mines. Retrieved from https://www.geologyontario.mines.gov.on.ca/mineral-inventory/MDI42G16SE00006

Orris, G.J., Seo, Y., Briggs, D.A., Dunlap, P., Cocker, M.D. (2018). *Global rare earth element occurrence database.* U.S. Geological Survey data release.

Pang, G., Shen, C., van den Hengel, A. (2019). Deep anomaly detection with deviation networks. *Proceedings of the 25th acm sigkdd international conference on knowledge discovery & data mining* (pp. 353–362).

Parker, B.J., Günter, S., Bedo, J. (2007). Stratification bias in low signal microarray studies. *BMC Bioinformatics*, *8*(1), 326, https://doi.org/10.1186/1471-2105-8-326

Parsa, M., & Carranza, E.J.M. (2021). Modulating the impacts of stochastic uncertainties linked to deposit locations in data-driven predictive mapping of mineral prospectivity. *Natural Resources Research*, *30*(5), 3081–3097, https://doi.org/10.1007/s11053-021-09891-9

Parsa, M., Lawley, C.J.M., Cawood, T., Martins, T., Cumani, R., Zhang, S.E., … Voinot, A. (2025). Pan-Canadian predictive modeling of lithium–cesium–tantalum pegmatites with deep learning and natural language processing. *Natural Resources Research*, *34*, 639–668, https://doi.org/10.1007/s11053-024-10438-x

Parsa, M., Lawley, C.J.M., Cumani, R., Schetselaar, E., Harris, J., Lentz, D.R., … Bourdeau, J.E. (2024). Predictive modeling of Canadian carbonatite-hosted REE +/– Nb deposits. *Natural Resources Research*, *33*(5), 1941–1965, https://doi.org/10.1007/s11053-024-10369-7

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., … Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, *12*, 2825–2830,

Percival, J.A., & West, G.F. (1994). The Kapuskasing uplift: A geological and geophysical synthesis. *Canadian Journal of Earth Sciences*, *31*(7), 1256–1286, https://doi.org/10.1139/e94-110

Ploton, P., Mortier, F., Réjou-Méchain, M., Barbier, N., Picard, N., Rossi, V., … Pélissier, R. (2020). Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nature Communications*, *11*, 4540, https://doi.org/10.1038/s41467-020-18321-y

Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., Guillera-Arroita, G., … Dormann, C.F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, *40*(8), 913–929, https://doi.org/10.1111/ecog.02881

Rukhlov, A.S., & Bell, K. (2010). Geochronology of carbonatites from the Canadian and Baltic Shields, and the Canadian Cordillera: Clues to mantle evolution. *Mineralogy and Petrology*, *98*(1), 11–54, https://doi.org/10.1007/s00710-009-0054-5

Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, *10*(3), e0118432, https://doi.org/10.1371/journal.pone.0118432

Simandl, G.J., & Paradis, S. (2018). Carbonatites: Related ore deposits, resources, footprint, and exploration methods. *Applied Earth Science*, *127*(4), 123–152, https://doi.org/10.1080/25726838.2018.1516935

Snyder, D.B., Savard, G., Kjarsgaard, B.A., Vaillancourt, A., Thurston, P.C., Ayer, J.A., Roots, E. (2021). Multidisciplinary modeling of mantle lithosphere structure within the Superior Craton, North America. *Geochemistry, Geophysics, Geosystems*, *22*(4), e2020GC009566, https://doi.org/10.1029/2020GC009566

Thomas, M.D., Ford, K.L., Keating, P. (2016). Review paper: Exploration geophysics for intrusion-hosted rare metals. *Geophysical Prospecting*, *64*(5), 1275–1304, https://doi.org/10.1111/1365-2478.12352

Valavi, R., Elith, J., Lahoz-Monfort, J.J., Guillera-Arroita, G. (2019). blockCV: An R package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. *Methods in Ecology and Evolution*, *10*(2), 225–232, https://doi.org/10.1111/2041-210X.13107

Verplanck, P.L., Van Gosen, B.S., Seal II, R.R., McCafferty, A.E. (2014). *A deposit model for carbonatite and peralkaline intrusion-related rare earth element deposits* (Scientific Investigations Report No. 2010-5070-J). U.S. Geological Survey.

Wang, Z.-Y., Fan, H.-R., Zhou, L., Yang, K.-F., She, H.-D. (2020). Carbonatite-related REE deposits: An overview. *Minerals*, *10*(11), 965, https://doi.org/10.3390/min10110965

Wheeler, J.O., Hoffman, P.F., Card, K.D., Davidson, A., Sanford, B.V., Okulitch, A.V., Roest, W.R. (1996). *Geological map of Canada* (Map No. 1860A). Geological Survey of Canada.

Woolley, A.R., & Bailey, D.K. (2012). The crucial role of lithospheric structure in the generation and release of carbonatites: Geological evidence. *Mineralogical Magazine*, *76*(2), 259–270, https://doi.org/10.1180/minmag.2012.076.2.02

Xiong, Y., & Zuo, R. (2016). Recognition of geochemical anomalies using a deep autoencoder network. *Computers & Geosciences*, *86*, 75–82, https://doi.org/10.1016/j.cageo.2015.10.006

Xiong, Y., & Zuo, R. (2018). GIS-based rare events logistic regression for mineral prospectivity mapping. *Computers & Geosciences*, *111*, 18–25, https://doi.org/10.1016/j.cageo.2017.10.005

Xiong, Y., & Zuo, R. (2021). A positive and unlabeled learning algorithm for mineral prospectivity mapping. *Computers & Geosciences*, *147*, 104667, https://doi.org/10.1016/j.cageo.2020.104667

Yaxley, G.M., Kjarsgaard, B.A., Jaques, A.L. (2021). Evolution of carbonatite magmas in the upper mantle and crust. *Elements*, *17*(5), 315–320, https://doi.org/10.2138/gselements.17.5.315

Yousefi, M., & Carranza, E.J.M. (2015). Prediction–area (P–A) plot and C–A fractal analysis to classify and evaluate evidential maps for mineral prospectivity modeling. *Computers & Geosciences*, *79*, 69–81, https://doi.org/10.1016/j.cageo.2015.03.007

Zuo, R., Kreuzer, O.P., Wang, J., Xiong, Y., Zhang, Z., Wang, Z. (2021). Uncertainties in GIS-based mineral prospectivity mapping: Key types, potential impacts and possible solutions. *Natural Resources Research*, *30*(5), 3059–3079, https://doi.org/10.1007/s11053-021-09871-z

Zuo, R., & Wang, Z. (2020). Effects of random negative training samples on mineral prospectivity mapping. *Natural Resources Research*, *29*(6), 3443–3455, https://doi.org/10.1007/s11053-020-09668-6